

11. Low Power Mode

11.1 Overview

The MCU has several functions for reducing power consumption, such as setting clock dividers, EBCLK output control, SDCLK output control, stopping modules, power gating control, selecting operating power control modes, and transitions to low power modes.

[Table 11.1](#) lists the specifications of functions to reduce power consumption. [Table 11.2](#) to [Table 11.4](#) lists the conditions to shift to low power modes, states of the CPU and peripheral modules, and the method for cancelling each mode. After a reset, the MCU enters the program execution state, but modules, except for the DMAC, DTC, and SRAM, do not operate.

Table 11.1 Specifications of the low-power mode functions

Parameter	Specification
Reducing power consumption by switching clock signals	The frequency division ratio can be selected independently for the system clock, peripheral module clock, external bus clock, and MRAM interface clock.* ¹
EBCLK output control	BCLK output or high-level output can be selected.
SDCLK output control	SDCLK output or high-level output can be selected.
Module-stop	Functions can be stopped independently for each peripheral module.
Power gating control	This function can be controlled the power state of the power domain. - Control the turning On/OFF for the power domain - Control the retention of specific circuits during power gating
Processor low power modes	- CPU Sleep mode - CPU Deep Sleep mode
Low power modes	- Software Standby mode*² - Deep Software Standby mode^{1, 2, 3}*²

Note 1. For details, see [section 9, Clock Generation Circuit](#).

Note 2. This mode is not supported in external VDD mode.

Table 11.2 Operating state of processor low power mode (1 of 2)

Item	CPU _n Sleep mode		CPU _n Deep Sleep mode	
	n = 0 (CPU0)	n = 1 (CPU1)	n = 0 (CPU0)	n = 1 (CPU1)
Transition condition	When [Condition 1] or [Condition 2] after set CPU0.SCR.SLEEPDEEP = 0 [Condition 1] - WFI or WFE instruction - A valid interrupt request cannot be accepted to CPU^{*5} (including a transition from the time WFI instruction is executed to the time the transition to Sleep mode is completed) - No event from CPU1 [Condition 2] - CPU0.SCR.SLEEPO NEXIT = 1 - Complete execution of all exception handlers - A valid interrupt request cannot be accepted to CPU^{*5} (including a transition from the time WFI instruction is executed to the time the transition to Sleep mode is completed) - No event from CPU1	When [Condition 1] or [Condition 2] after set CPU1.SCR.SLEEPDEEP = 0 [Condition 1] - WFI or WFE instruction - A valid interrupt request cannot be accepted to CPU^{*5} (including a transition from the time WFI instruction is executed to the time the transition to Sleep mode is completed) - No event from CPU0 [Condition 2] - CPU1.SCR.SLEEPO NEXIT = 1 - Complete execution of all exception handlers - A valid interrupt request cannot be accepted to CPU^{*5} (including a transition from the time WFI instruction is executed to the time the transition to Sleep mode is completed) - No event from CPU0	When [Condition 1] or [Condition 2] after set CPU0.SCR.SLEEPDEEP = 1 [Condition 1] - WFI instruction - A valid interrupt request cannot be accepted to CPU^{*5} (including a transition from the time WFI instruction is executed to the time the transition to Deep Sleep mode is completed) [Condition 2] - CPU0.SCR.SLEEPO NEXIT = 1 - Complete execution of all exception handlers - A valid interrupt request cannot be accepted to CPU^{*5} (including a transition from the time WFI instruction is executed to the time the transition to Deep Sleep mode is completed)	When [Condition 1] or [Condition 2] after set CPU1.SCR.SLEEPDEEP = 1 [Condition 1] - WFI instruction - A valid interrupt request cannot be accepted to CPU^{*5} (including a transition from the time WFI instruction is executed to the time the transition to Deep Sleep mode is completed) [Condition 2] - CPU1.SCR.SLEEPO NEXIT = 1 - Complete execution of all exception handlers - A valid interrupt request cannot be accepted to CPU^{*5} (including a transition from the time WFI instruction is executed to the time the transition to Deep Sleep mode is completed)
Canceling method	All interrupts. Any reset available in the mode.	All interrupts. Any reset available in the mode.	Interrupts shown in Table 14.5 Any reset available in the mode.	Interrupts shown in Table 14.5 Any reset available in the mode.
State after cancellation by an interrupt	Program execution state	Program execution state	Program execution state	Program execution state
State after cancellation by a reset	Reset state	Reset state	Reset state	Reset state
CPU0	Stop (Retained)	—	Stop (Retained)	—
CPU0 TCM (SRAM)	—	—	Stop (Retained) ^{*4}	—
DMA Controller 0 (DMAC0)	Selectable	—	Selectable	—
Data Transfer Controller 0 (DTC0)	Selectable	—	Selectable	—
Watchdog Timer 0 (WDT0)	Selectable ^{*1}	—	Selectable ^{*1}	—
CPU1	—	Stop (Retained)	—	Stop (Retained)
DMA Controller 1 (DMAC1)	—	Selectable	—	Selectable
Data Transfer Controller 1 (DTC1)	—	Selectable	—	Selectable
Watchdog Timer 1 (WDT1)	—	Selectable ^{*1}	—	Selectable ^{*1}

Table 11.2 Operating state of processor low power mode (2 of 2)

Item	CPU _n Sleep mode		CPU _n Deep Sleep mode	
	n = 0 (CPU0)	n = 1 (CPU1)	n = 0 (CPU0)	n = 1 (CPU1)
Independent Watchdog Timer (IWDG)	Selectable* ¹	Selectable* ¹	Selectable* ¹	Selectable* ¹
ARM Debug function	Stop* ²	Stop* ²	Stop* ²	Stop* ²
Trace function	Stop* ³	Stop* ³	Stop* ³	Stop* ³
Other peripheral modules	—	—	—	—

Note: Selectable means that operating or not operating can be selected by setting the control registers before entering Processor Low Power Mode.

Stop (Retained) means that the contents of the internal registers are retained but the operations are suspended.

Stop (Undefined) means that the contents of the internal registers are undefined and power to the internal circuit is cut off.

“—” means that the operation is not affected.

Note 1. In IWDG, operating or stopping is selected by setting the IWDG Stop Control bit (IWDGSTOPCTL) in Option Function Select register 0 (OFS0) in IWDG auto start mode. Also, operating or stopping is selected by setting the IWDG.IWDGSTOPR.SLCSTP bit in IWDG register start mode.

In WDT, operating or stopping is selected by setting the WDT Stop Control bit (WDTSTOPCTL) in Option Function Select Register 0 (OFS0) in WDT auto start mode. Also, operating or stopping is selected by setting the WDT.WDTSTOPR.SLCSTP bit in WDT register start mode.

Note 2. This function is operating if debugger is connecting (DBGSTR.CDBGPWRUPREQ = 1 and SYOCDRC.DBGEN = 1) before entering this mode.

Note 3. This function is operating if debugger is connecting (DBGSTR.CDBGPWRUPREQ = 1 and SYOCDRC.DBGEN = 1) and TRCKCR.TRCKEN = 1 before entering this mode.

Note 4. If PDRAMSCR1.RKEEP0 bit is set to 0, the contents of the CPU0 TCM is not retained.

Note 5. Valid interrupt requests are any interrupt or exception that are not masked by the priority level of current exception and the priority level set by BASEPRI. In addition, if the interrupt request is based on IELSRn, the interrupt must be enabled by NVIC_ISERN.

Table 11.3 Operating state of each low power mode (1 of 3)

Item	Software Standby mode (SSTBY)	Deep Software Standby mode (DSTBY)		
	SSTBY	DSTBY 1	DSTBY 2	DSTBY 3
Transition condition	When [Condition 1] or [Condition 2] after set LPSCR and CPU _n .SCR.SLEEPDEEP = 1 (n = 0, 1) [Condition 1] - WFI instruction - A valid interrupt request cannot be accepted to CPU*²⁴ (including a transition from the time WFI instruction is executed to the time the transition to Software Standby mode is completed) [Condition 2] - CPU_n.SCR.SLEEPONEXIT = 1 - Complete execution of all exception handlers - A valid interrupt request cannot be accepted to CPU*²⁴ (including a transition from the time WFI instruction is executed to the time the transition to Software Standby mode is completed)	When [Condition 1] or [Condition 2] after set LPSCR and CPU _n .SCR.SLEEPDEEP = 1 (n = 0, 1) [Condition 1] - WFI instruction - A valid interrupt request cannot be accepted to CPU*²⁴ (including a transition from the time WFI instruction is executed to the time the transition to Software Standby mode is completed) [Condition 2] - CPU_n.SCR.SLEEPONEXIT = 1 - Complete execution of all exception handlers - A valid interrupt request cannot be accepted to CPU*²⁴ (including a transition from the time WFI instruction is executed to the time the transition to Software Standby mode is completed)		
Canceling method	Interrupts shown in Table 11.4 . Any reset available in the mode.	Interrupts shown in Table 11.4 . Any reset available in the mode.		
State after cancellation by an interrupt	Program execution state	Reset state		
State after cancellation by a reset	Reset state	Reset state		
Main clock oscillator	Selectable* ¹⁰	Stop		
Sub-clock oscillator	Selectable	Selectable		

Table 11.3 Operating state of each low power mode (2 of 3)

	Software Standby mode (SSTBY)	Deep Software Standby mode (DSTBY)		
Item	SSTBY	DSTBY 1	DSTBY 2	DSTBY 3
High-speed on-chip oscillator	Selectable ^{*11}	Stop		
Middle-speed on-chip oscillator	Selectable ^{*18}	Stop ^{*18}		
Low-speed on-chip oscillator	Selectable ^{*2}	Selectable ^{*2}	Stop	
PLL1	Stop	Stop		
PLL2	Stop	Stop		
Oscillation stop detection function	Selectable ^{*12}	Stop		
Sub-clock Oscillation stop detection function	Selectable	Selectable		
Clock/buzzer output function	Selectable ^{*3}	Stop (Undefined)		
External Bus (EBCLK)	Stop (Retained)	Stop (Retained)		
CPU	Stop (Retained)	Stop (Undefined)		
TCM (SRAM)	Stop (Retained) ^{*14}	Stop (Undefined)		
User SRAM	Stop (Retained) ^{*13}	Stop (Undefined)		
Backup register	Stop (Retained)	Stop (Retained)		
MRAM	Stop (Retained)	Stop (Retained)		
Independent Watchdog Timer (IWDT)	Selectable ^{*1}	Selectable ^{*1}	Stop (Undefined)	
ARM Debug function	Stop ^{*15}	Stop ^{*15}		
Trace function	Stop ^{*16}	Stop ^{*16}		
Layer3 Ethernet Switch Module (ESWM)	Stop (Undefined)	Stop (Undefined)		
EtherCAT Slave Controller (ESC)	Stop (Retained) ^{*23}	Stop (Undefined)	Stop (Undefined)	
USB 2.0 Full-Speed (USBFS)	Stop (Retained) ^{*22} Detection of USB resumption is possible.	Stop (Retained) Detection of USB resumption is possible.	Stop (Undefined)	
Realtime Clock (RTC)	Selectable	Selectable	Selectable ^{*4}	
CAN with Flexible Data-rate (CANFD)	Stop (Undefined)	Stop (Undefined)		
SD/MMC Host Interface (SDHI)	Stop (Undefined)	Stop (Undefined)		
Ultra low power Timer (ULPT)	Selectable	Selectable	Stop (Undefined)	
Asynchronous General Purpose Timer (AGT)	Selectable ^{*5}	Stop (Undefined)		
I2C Bus Interface (IIC0)	Selectable ^{*6}	Stop (Undefined)		
I2C Bus Interface (IICn, n ≥1)	Stop (Retained)	Stop (Undefined)		
I3C Bus Interface (I3C)	Selectable ^{*21}	Stop (Undefined)		
12-Bit D/A Converter (DAC12)	Selectable	Selectable		

Table 11.3 Operating state of each low power mode (3 of 3)

Item	Software Standby mode (SSTBY)	Deep Software Standby mode (DSTBY)		
	SSTBY	DSTBY 1	DSTBY 2	DSTBY 3
Temperature sensor (TSN)	Selectable ^{*20}	Selectable ^{*20}		
High-Speed Analog Comparator 0 (ACMPHS0)	Selectable	Stop (Undefined)		
High-Speed Analog Comparator n (ACMPHSn, n≥1)	Selectable ^{*19}	Stop (Undefined)		
IRQn (n ≥ 0) pin interrupt	Selectable	Stop (Undefined)		
NMI, IRQn-DS (n ≥ 0) pin interrupt	Selectable	Selectable		
Programmable Voltage Detect (PVD)	Selectable	Selectable ^{*7}	Selectable ^{*7}	Stop (Undefined) ^{*7}
VCC voltage drop detection for VBATT	Selectable	Selectable ^{*17}	Selectable ^{*17}	Stop (Undefined)
Power-on reset circuit	Operating	Operating	Operating	Operating ^{*8}
CVM reset circuit	Selectable	Stop	Stop	Stop
Other peripheral modules	Stop (Retained)	Stop (Undefined)		
I/O Ports	Retained ^{*9}	Retained ^{*9}		

Note: **Selectable** means that operating or not operating can be selected by setting the control registers before low power mode.

Stop (Retained) means that the contents of the internal registers are retained but the operations are suspended.

Stop (Undefined) means that the contents of the internal registers are undefined and power to the internal circuit is cut off. After canceling from each low power mode, the internal registers are initialized.

- Note 1. In IWDT, operating or stopping is selected by setting the IWDT Stop Control bit (IWDTSTPCTL) in Option Function Select register 0 (OFS0) in IWDT auto start mode. Also, operating or stopping is selected by setting the IWDT.IWDTCSSTPR.SLCSTP bit in IWDT register start mode.
- Note 2. If IWDT is not used and LOCOCR.LCSTP = 0, LOCO is not stopped in Software Standby mode or Deep Software Standby mode 1. If IWDT is used and IWDT Stop Control bit is 0 (OFS.IWDTSTPCTL = 0 or IWDT.IWDTCSSTPR.SLCSTP = 0), LOCO is not stopped regardless the value of LOCOCR.LCSTP in Software Standby mode or Deep Software Standby mode 1. If IWDT is used and IWDT Stop Control bit is 1 (OFS.IWDTSTPCTL = 1 or IWDT.IWDTCSSTPR.SLCSTP = 1), LOCO is not stopped with LOCOCR.LCSTP = 0 in Software Standby mode or Deep Software Standby mode 1. In other cases, LOCO is stopped in Software Standby mode or Deep Software Standby mode 1.
- Note 3. Stopped when the clock output source select bits (CKOCR.CKSEL[2:0]) are set to a value other than 010 (LOCO) and 100 (SOSC).
- Note 4. Only sub clock oscillator can be selected as the count source clock for RTC. When the RCR4.RCKSEL bit set to 1 (LOCO), the LPSCR must set to Deep Software Standby mode 1 before entering Deep Software Standby mode.
- Note 5. AGT0 operation is possible when 100 (AGTLCLK) or 110 (AGTSLCK) is selected by the AGT0.AGTMR1.TCK[2:0] bits. AGT1 operation is possible when 100 (AGTLCLK), 110 (AGTSLCK) or 101 (Underflow event signal from AGT0) is selected by the AGT1.AGTMR1.TCK[2:0] bits.
- Note 6. Only IIC0 wakeup interrupt is available.
- Note 7. When using PVD in Deep Software Standby mode, LPSCR must be set to DSTBY1 or DSTBY2 before entering Deep Software Standby mode.
- Note 8. When the MCU enters Deep Software Standby mode 3, the PVD circuit stops.
- Note 9. For the address bus and bus control signals (For SRAM: [A00 to A23, CS0 to CS7, RD, WR0 to WR3, WR, BC0 to BC3 and ALE], and for SDRAM: [A00 to A16, DQM0 to DQM3, SDCS, RAS, CAS, WE and CKE]), keeping the output state or changing to the high-impedance state can be selected by SBYCR.OPE bit.
- Note 10. When MOSCSCR.MOSCSOKP = 1 while MOSC is oscillated, MOSC continues to oscillate in Software Standby mode.
- Note 11. When HOCOSCR.HOCOSOKP = 1 while HOCO is oscillated, HOCO continues to oscillate in Software Standby mode.
- Note 12. The function is depending on the setting value of the MOSCSCR.MOSCSOKP bit.
- Note 13. If PDRAMSCR0.RKEEPn bit is set to 0, the contents of the target User SRAM is not retained.
- Note 14. If PDRAMSCR1.RKEEPn bit is set to 0, the contents of the target TCM is not retained.
- Note 15. This function is operating if debugger is connecting before entering this mode.
- Note 16. This function is operating if debugger is connecting and TRCKCR.TRCKEN = 1 before entering this mode.
- Note 17. When the MCU enters the Deep Software Standby mode 1 or 2 with that OFS1.PVDAS = 0 and OFS1.PVDLPSEL = 0, the battery power supply switch is controlled by the voltage monitor 0 which has lower power consumption than VCC voltage drop detection circuit dedicated for VBATT. When the MCU enters the Deep Software Standby mode 1 or 2 with that OFS1.PVDLPSEL = 1, the battery power supply switch is controlled by the VCC voltage drop detection circuit dedicated for VBATT. See [section 12, Battery Backup Function](#) and [section 60, Electrical Characteristics](#) for details.
- Note 18. When the on-chip debugger function is valid, MOCO does not stop in this mode.

Note 19. Only VCOOUT function is permitted. The VCOOUT pin operates when ACMPHS uses no digital filter. For details on digital filter, see [section 55, High-Speed Analog Comparator \(ACMPHS\)](#).

Note 20. Only temperature monitor reset function can be operated.

Note 21. Only I3C wakeup interrupt is available.

Note 22. FIFO buffer is not retained.

Note 23. User RAM and Process Data RAM are not retained.

Note 24. Valid interrupt requests are any interrupt or exception that are not masked by the priority level of current exception and the priority level set by BASEPRI. In addition, if the interrupt request is based on IELSRn, the interrupt must be enabled by NVIC_ISERn.

Table 11.4 Interrupt source for canceling Software Standby and Deep Software Standby modes

Interrupt source	Name	Software Standby mode	Deep Software Standby mode		
		SSTBY	DSTBY1	DSTBY2	DSTBY3
NMI		Yes	Yes	Yes	Yes
Port	PORT_IRQn	Yes	No	No	No
	PORT_IRQn-DS	Yes	Yes	Yes	Yes
PVD	PVD_PVDm (m = 1, 2)	Yes	Yes	Yes	No
IWDT	IWDT_NMIUNDF	Yes	Yes	No	No
USBFS	USBFS_USBR	Yes	Yes	No	No
RTC	RTC_ALM	Yes	Yes	Yes	Yes
	RTC_PRD	Yes	Yes	Yes	Yes
ULPT0	ULPT0_ULPTI	Yes	Yes	No	No
	ULPT0_ULPTCMAI	Yes	No	No	No
	ULPT0_ULPTCMBI	Yes	No	No	No
ULPT1	ULPT1_ULPTI	Yes	Yes	No	No
	ULPT1_ULPTCMAI	Yes	No	No	No
	ULPT1_ULPTCMBI	Yes	No	No	No
AGT1	AGT1_AGTI	Yes	No	No	No
	AGT1_AGTCMAI	Yes	No	No	No
	AGT1_AGTCMBI	Yes	No	No	No
ACMPHS0	ACMP_HS0	Yes	No	No	No
IIC0	IIC0_WUI	Yes	No	No	No
I3C	I3C_WU	Yes	No	No	No
Sub-clock Oscillation stop detection function	SOSC_STOP	Yes	Yes	Yes	Yes
BBF (Battery Backup Function)	VBATT_TADI	Yes	Yes	Yes	Yes

[Figure 11.1](#) lists the low power mode transitions.

For detail, see [section 11.6. Low Power Modes](#).

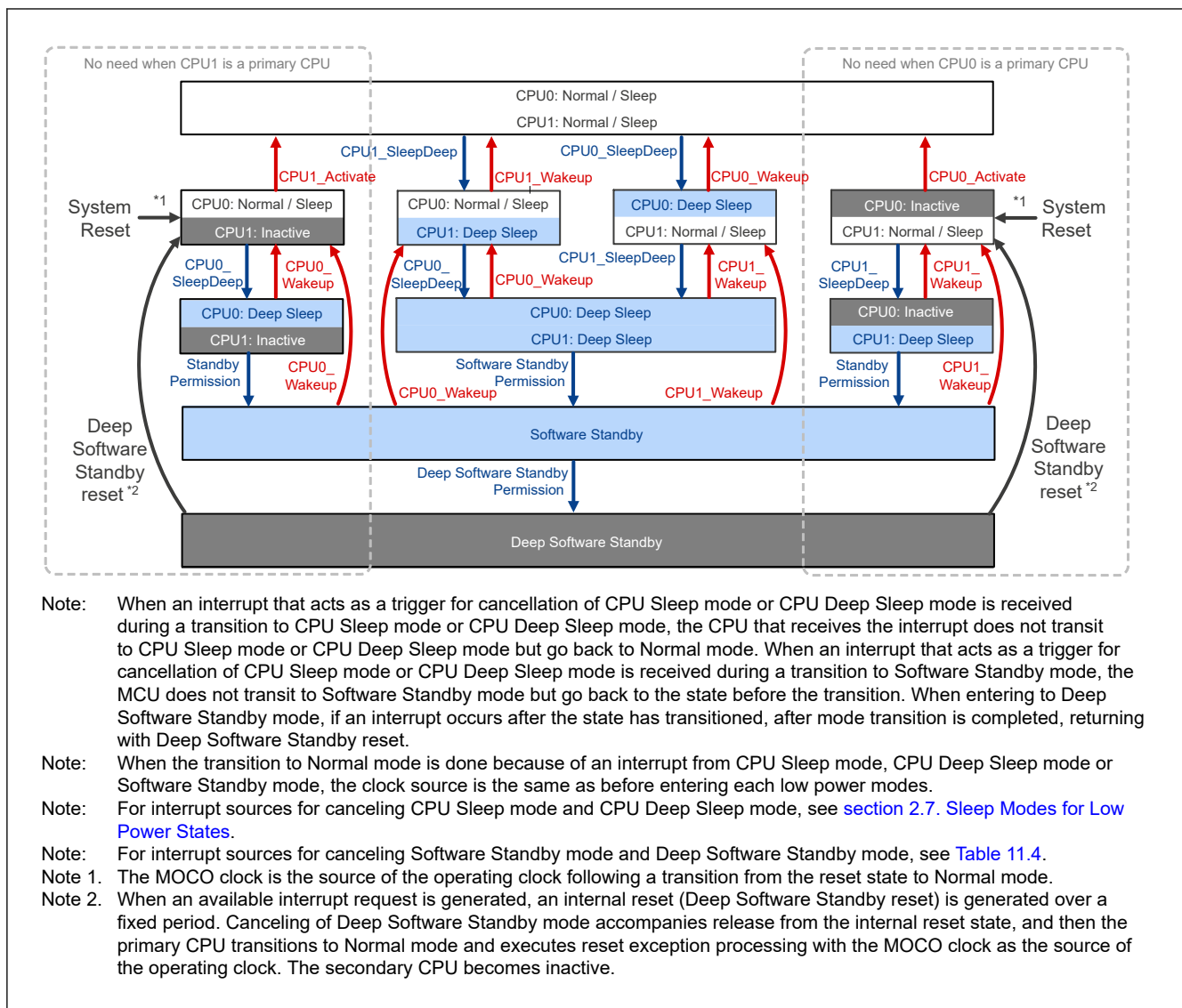


Figure 11.1 Low power mode transitions

11.2 Register Descriptions

11.2.1 LPMSAR : Low Power Mode Security Attribution Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x3C8

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	NONS EC21	—	NONS EC19	NONS EC18	NONS EC17	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	NONS EC8	—	—	—	NONS EC4	—	NONS EC2	NONS EC1	NONS EC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	NONSEC0	Non-secure Attribute bit 0 Target register: OPCCR 0: Secure 1: Non-secure	R/W
1	NONSEC1	Non-secure Attribute bit 1 Target register: LPSCR 0: Secure 1: Non-secure	R/W
2	NONSEC2	Non-secure Attribute bit 2 Target register: SBYCR, SSCR1, SVSCR 0: Secure 1: Non-secure	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
4	NONSEC4	Non-secure Attribute bit 4 Target register: VSCR 0: Secure 1: Non-secure	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W
8	NONSEC8	Non-secure Attribute bit 8 Target register: DPSBYCR 0: Secure 1: Non-secure	R/W
16:9	—	These bits are read as 0. The write value should be 0.	R/W
17	NONSEC17	Non-secure Attribute bit 17 Target register: HOCOLDOCR 0: Secure 1: Non-secure	R/W
18	NONSEC18	Non-secure Attribute bit 18 Target register: PLL1LDOCR 0: Secure 1: Non-secure	R/W
19	NONSEC19	Non-secure Attribute bit 19 Target register: PLL2LDOCR 0: Secure 1: Non-secure	R/W
20	—	This bit is read as 0. The write value should be 0.	R/W
21	NONSEC21	Non-secure Attribute bit 21 Target register: LVOCR, MWMCR 0: Secure 1: Non-secure	R/W
31:22	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-1, P-TYPE-1

The LPMSAR register controls the secure attribute of Low Power Mode registers.

NONSEC0 bit (Non-secure Attribute bit 0)

This bit controls the security attribute of OPCCR.

NONSEC1 bit (Non-secure Attribute bit 1)

This bit controls the security attribute of LPSCR.

NONSEC2 bit (Non-secure Attribute bit 2)

This bit controls the security attribute of SBYCR, SSCR1, SVSCR.

NONSEC4 bit (Non-secure Attribute bit 4)

This bit controls the security attribute of VSCR.

NONSEC8 bit (Non-secure Attribute bit 8)

This bit controls the security attribute of DPSBYCR.

NONSEC17 bit (Non-secure Attribute bit 17)

This bit controls the security attribute of HOCOLDOCR.

NONSEC18 bit (Non-secure Attribute bit 18)

This bit controls the security attribute of PLL1LDOCR.

NONSEC19 bit (Non-secure Attribute bit 19)

This bit controls the security attribute of PLL2LDOCR.

NONSEC21 bit (Non-secure Attribute bit 21)

This bit controls the security attribute of LVOCR and MWMCR.

11.2.2 PGCSAR : Power Gating Control Security Attribution Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x3D8

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	NONS EC7	NONS EC6	NONS EC5	NONS EC4	NONS EC3	—	—	NONS EC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	NONSEC0	Non-secure Attribute bit 0 Target register: PGSCR 0: Secure 1: Non-secure	R/W
2:1	—	These bits are read as 0. The write value should be 0.	R/W
3	NONSEC3	Non-secure Attribute bit 3 Target register: PSSTCR2 0: Secure 1: Non-secure	R/W
4	NONSEC4	Non-secure Attribute bit 4 Target register: PSSTCR3 0: Secure 1: Non-secure	R/W
5	NONSEC5	Non-secure Attribute bit 5 Target register: PSSTCR4 0: Secure 1: Non-secure	R/W
6	NONSEC6	Non-secure Attribute bit 6 Target register: PSSTCR5, PDCTRESWM 0: Secure 1: Non-secure	R/W
7	NONSEC7	Non-secure Attribute bit 7 Target register: CPUDSCR 0: Secure 1: Non-secure	R/W

Bit	Symbol	Function	R/W
31:8	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-1, P-TYPE-1

The PGCSAR register controls the secure attribute of power gating registers.

NONSEC0 bit (Non-secure Attribute bit 0)

This bit controls the security attribute of PGSCR.

NONSEC3 bit (Non-secure Attribute bit 3)

This bit controls the security attribute of PSSTCR2.

NONSEC4 bit (Non-secure Attribute bit 4)

This bit controls the security attribute of PSSTCR3.

NONSEC5 bit (Non-secure Attribute bit 5)

This bit controls the security attribute of PSSTCR4.

NONSEC6 bit (Non-secure Attribute bit 6)

This bit controls the security attribute of PSSTCR5, PDCTRESWM.

NONSEC7 bit (Non-secure Attribute bit 7)

This bit controls the security attribute of CPUDSCR.

11.2.3 DPFSAR : Deep Software Standby Interrupt Factor Security Attribution Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x3E0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	DPFS A31	DPFS A30	DPFS A29	—	DPFS A27	DPFS A26	—	DPFS A24	—	—	—	DPFS A20	DPFS A19	DPFS A18	DPFS A17	DPFS A16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DPFSA								DPFSA							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	DPFSA7 to DPFSA0	Deep Software Standby Interrupt Factor Security Attribute bit n (n = 0 to 7) Target register: DPSIER0.bn, DPSIFR0.bn, DPSIEGR0.bn (n = 0 to 7) Target factor : IRQn-DS Pin (n = 0 to 7) 0: Secure 1: Non-secure	R/W
15:8	DPFSA15 to DPFSA8	Deep Software Standby Interrupt Factor Security Attribute bit n (n = 8 to 15) Target register: DPSIER1.bn, DPSIFR1.bn, DPSIEGR1.bn (n = 0 to 7) Target factor : IRQn-DS Pin (n = 8 to 15) 0: Secure 1: Non-secure	R/W
16	DPFSA16	Deep Software Standby Interrupt Factor Security Attribute bit 16 Target register: DPSIER2.b0, DPSIFR2.b0, DPSIEGR2.b0 Target factor: PVD1 0: Secure 1: Non-secure	R/W

Bit	Symbol	Function	R/W
17	DPFSA17	Deep Software Standby Interrupt Factor Security Attribute bit 17 Target register: DPSIER2.b1, DPSIFR2.b1, DPSIEGR2.b1 Target factor: PVD2 0: Secure 1: Non-secure	R/W
18	DPFSA18	Deep Software Standby Interrupt Factor Security Attribute bit 18 Target register: DPSIER2.b2, DPSIFR2.b2 Target factor: RTC Interval 0: Secure 1: Non-secure	R/W
19	DPFSA19	Deep Software Standby Interrupt Factor Security Attribute bit 19 Target register: DPSIER2.b3, DPSIFR2.b3 Target factor: RTC Alarm 0: Secure 1: Non-secure	R/W
20	DPFSA20	Deep Software Standby Interrupt Factor Security Attribute bit 20 Target register: DPSIER2.b4, DPSIFR2.b4, DPSIEGR2.b4 Target factor: NMI Pin 0: Secure 1: Non-secure	R/W
23:21	—	These bits are read as 0. The write value should be 0.	R/W
24	DPFSA24	Deep Software Standby Interrupt Factor Security Attribute bit 24 Target register: DPSIER3.b0, DPSIFR3.b0 Target factor: USBFS Suspend/Resume 0: Secure 1: Non-secure	R/W
25	—	This bit is read as 0. The write value should be 0.	R/W
26	DPFSA26	Deep Software Standby Interrupt Factor Security Attribute bit 26 Target register: DPSIER3.b2, DPSIFR3.b2 Target factor: ULPT0 0: Secure 1: Non-secure	R/W
27	DPFSA27	Deep Software Standby Interrupt Factor Security Attribute bit 27 Target register: DPSIER3.b3, DPSIFR3.b3 Target factor: ULPT1 0: Secure 1: Non-secure	R/W
28	—	This bit is read as 0. The write value should be 0.	R/W
29	DPFSA29	Deep Software Standby Interrupt Factor Security Attribute bit 29 Target register: DPSIER3.b5, DPSIFR3.b5 Target factor: IWDG Overflow 0: Secure 1: Non-secure	R/W
30	DPFSA30	Deep Software Standby Interrupt Factor Security Attribute bit 30 Target register: DPSIER3.b6, DPSIFR3.b6 Target factor: Sub-clock Oscillation stop detection 0: Secure 1: Non-secure	R/W
31	DPFSA31	Deep Software Standby Interrupt Factor Security Attribute bit 31 Target register: DPSIER3.b7, DPSIFR3.b7 Target factor: Tamper Detection 0: Secure 1: Non-secure	R/W

Note: S-TYPE-1, P-TYPE-1

The DPFSAR register controls the secure attribute of Deep Software Standby Interrupt Factor control registers.

DPFSA bits (Deep Software Standby Interrupt Factor Security Attribute bit n (n = 0 to 7))

This bit controls the security attribute of DPSIER0.bn, DPSIFR0.bn, DPSIEGR0.bn (n = 0 to 7).

Target factor is IRQn-DS Pin (n = 0 to 7).

DPFSA bit (Deep Software Standby Interrupt Factor Security Attribute bit n (n = 8 to 15))

This bit controls the security attribute of DPSIER1.bn, DPSIFR1.bn, DPSIEGR1.bn (n = 0 to 7).

Target factor is IRQn-DS Pin (n = 8 to 15).

DPFSA16 bit (Deep Software Standby Interrupt Factor Security Attribute bit 16)

This bit controls the security attribute of DPSIER2.b0, DPSIFR2.b0, DPSIEGR2.b0.

Target factor is PVD1.

DPFSA17 bit (Deep Software Standby Interrupt Factor Security Attribute bit 17)

This bit controls the security attribute of DPSIER2.b1, DPSIFR2.b1, DPSIEGR2.b1.

Target factor is PVD2.

DPFSA18 bit (Deep Software Standby Interrupt Factor Security Attribute bit 18)

This bit controls the security attribute of DPSIER2.b2, DPSIFR2.b2.

Target factor is RTC Interval.

DPFSA19 bit (Deep Software Standby Interrupt Factor Security Attribute bit 19)

This bit controls the security attribute of DPSIER2.b3, DPSIFR2.b3.

Target factor is RTC Alarm.

DPFSA20 bit (Deep Software Standby Interrupt Factor Security Attribute bit 20)

This bit controls the security attribute of DPSIER2.b4, DPSIFR2.b4, DPSIEGR2.b4.

Target factor is NMI Pin.

DPFSA24 bit (Deep Software Standby Interrupt Factor Security Attribute bit 24)

This bit controls the security attribute of DPSIER3.b0, DPSIFR3.b0.

Target factor is USBFS Suspend/Resume.

DPFSA26 bit (Deep Software Standby Interrupt Factor Security Attribute bit 26)

This bit controls the security attribute of DPSIER3.b2, DPSIFR3.b2.

Target factor is ULPT0.

DPFSA27 bit (Deep Software Standby Interrupt Factor Security Attribute bit 27)

This bit controls the security attribute of DPSIER3.b3, DPSIFR3.b3.

Target factor is ULPT1.

DPFSA29 bit (Deep Software Standby Interrupt Factor Security Attribute bit 29)

This bit controls the security attribute of DPSIER3.b5, DPSIFR3.b5.

Target factor is IWDT Overflow.

DPFSA30 bit (Deep Software Standby Interrupt Factor Security Attribute bit 30)

This bit controls the security attribute of DPSIER3.b6, DPSIFR3.b6.

Target factor is Sub-clock Oscillation stop detection.

DPFSA31 bit (Deep Software Standby Interrupt Factor Security Attribute bit 31)

This bit controls the security attribute of DPSIER3.b7, DPSIFR3.b7.

Target factor is Tamper Detection.

11.2.4 DPFSAR1 : Deep Software Standby Interrupt Factor Security Attribution Register 1

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x3E8

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DPFSA[15:8]								DPFSA[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	DPFSA7 to DPFSA0	Deep Software Standby Interrupt Factor Security Attribute bit n (n = 0 to 7) Target register: DPSIER4.bn, DPSIFR4.bn, DPSIEGR3.bn (n = 0 to 7) Target factor: IRQn-DS Pin (n = 16 to 23) 0: Secure 1: Non-secure	R/W
15:8	DPFSA15 to DPFSA8	Deep Software Standby Interrupt Factor Security Attribute bit n (n = 8 to 15) Target register: DPSIER5.bn, DPSIFR5.bn, DPSIEGR4.bn (n = 0 to 7) Target factor: IRQn-DS Pin (n = 24 to 31) 0: Secure 1: Non-secure	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-1, P-TYPE-1

The DPFSAR1 register controls the secure attribute of Deep Software Standby Interrupt Factor control registers.

DPFSA bit (Deep Software Standby Interrupt Factor Security Attribute bit n (n = 0 to 7))

This bit controls the security attribute of DPSIER4.bn, DPSIFR4.bn, DPSIEGR3.bn (n = 0 to 7).

Target factor is IRQn-DS Pin (n = 16 to 23).

DPFSA bit (Deep Software Standby Interrupt Factor Security Attribute bit n (n = 8 to 15))

This bit controls the security attribute of DPSIER5.bn, DPSIFR5.bn, DPSIEGR4.bn.

Target factor is IRQn-DS Pin (n = 24 to 31).

11.2.5 RSCSAR : RAM Standby Control Security Attribution Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x3E4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	RSCSA[17:16]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	RSCSA[12:0]												
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
12:0	RSCSA[12:0]	RAM Standby Control Security Attribute bit n (n = 0 to 12) Target register: PDRAMSCR0.bn (n = 0 to 12) 0: Secure 1: Non-secure	R/W
15:13	—	These bits are read as 0. The write value should be 0.	R/W
17:16	RSCSA[17:16]	RAM Standby Control Security Attribute bit n (n = 16 to 17) Target register: PDRAMSCR1.bn (n = 0 to 1) 0: Secure 1: Non-secure	R/W
31:18	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-1, P-TYPE-1

The RSCSAR register controls the secure attribute of RAM Standby Control registers.

RSCSA[12:0] bit (RAM Standby Control Security Attribute bit n (n = 0 to 12))

This bit controls the security attribute of PDRAMSCR0.bn (n = 0 to 12).

RSCSA[17:16] bit (RAM Standby Control Security Attribute bit n (n = 16 to 17))

This bit controls the security attribute of PDRAMSCR1.bn (n = 0 to 1).

11.2.6 MSTPCRA : Module Stop Control Register A

Base address: MSTP = 0x4020_3000
MSTP_NS = 0x5020_3000

Offset address: 0x00

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	MSTP A23	MSTP A22	—	—	—	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	0	0	1	1	1	1	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	MSTP A3	MSTP A2	MSTP A1	MSTP A0
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0

Bit	Symbol	Function	R/W
0	MSTPA0	SRAM0 Module Stop*2 Target module: SRAM0 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
1	MSTPA1	SRAM1 Module Stop*2 Target module: SRAM1 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
2	MSTPA2	SRAM2 Module Stop*2 Target module: SRAM2 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
3	MSTPA3	SRAM3 Module Stop*2 Target module: SRAM3 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
21:4	—	These bits are read as 1. The write value should be 1.	R/W

Bit	Symbol	Function	R/W
22	MSTPA22	DMA Controller 0 and Data Transfer Controller 0 Module Stop* ¹ Target modules: DMAC0, DTC0 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
23	MSTPA23	DMA Controller 1 and Data Transfer Controller 1 Module Stop* ¹ Target module: DMAC1, DTC1 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
31:24	—	These bits are read as 1. The write value should be 1.	R/W

Note: S-TYPE-4, P-TYPE-2

Note 1. When rewriting the MSTPAi bit from 0 to 1, disable the DMAC and DTC before setting MSTPAi bit (i = 22, 23).

Note 2. When changing the value of this bit, only execute subsequent instructions after reading this bit to check that the value was updated.

11.2.7 MSTPCRB : Module Stop Control Register B

Base address: MSTP = 0x4020_3000
MSTP_NS = 0x5020_3000

Offset address: 0x04

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	MSTP B31	MSTP B30	MSTP B29	MSTP B28	MSTP B27	MSTP B26	MSTP B25	MSTP B24	MSTP B23	MSTP B22	—	—	MSTP B19	MSTP B18	MSTP B17	MSTP B16
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	MSTP B11	—	MSTP B9	MSTP B8	MSTP B7	—	—	MSTP B4	—	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
3:0	—	These bits are read as 1. The write value should be 1.	R/W
4	MSTPB4	I3C Bus Interface Module Stop* ¹ Target module: I3C 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
6:5	—	These bits are read as 1. The write value should be 1.	R/W
7	MSTPB7	IIC Bus Interface 2 Module Stop Target module: IIC2 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
8	MSTPB8	IIC Bus Interface 1 Module Stop Target module: IIC1 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
9	MSTPB9	IIC Bus Interface 0 Module Stop Target module: IIC0 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
10	—	This bit is read as 1. The write value should be 1.	R/W
11	MSTPB11	Universal Serial Bus 2.0 FS Interface 0 Module Stop* ² Target module: USBFS0 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
15:12	—	These bits are read as 1. The write value should be 1.	R/W

Bit	Symbol	Function	R/W
16	MSTPB16	Octal Serial Peripheral Interface 0 and Decryption On The Fly 0 Module Stop* ³ Target module: OSPI0 and DOTF0 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
17	MSTPB17	Octal Serial Peripheral Interface 1 and Decryption On The Fly 1 Module Stop* ³ Target module: OSPI1 and DOTF1 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
18	MSTPB18	Serial Peripheral Interface 1 Module Stop* ⁴ Target module: SPI1 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
19	MSTPB19	Serial Peripheral Interface 0 Module Stop* ⁴ Target module: SPI0 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
21:20	—	These bits are read as 1. The write value should be 1.	R/W
22	MSTPB22	Serial Communication Interface 9 Module Stop* ⁵ Target module: SCI9 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
23	MSTPB23	Serial Communication Interface 8 Module Stop* ⁵ Target module: SCI8 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
24	MSTPB24	Serial Communication Interface 7 Module Stop* ⁵ Target module: SCI7 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
25	MSTPB25	Serial Communication Interface 6 Module Stop* ⁵ Target module: SCI6 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
26	MSTPB26	Serial Communication Interface 5 Module Stop* ⁵ Target module: SCI5 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
27	MSTPB27	Serial Communication Interface 4 Module Stop* ⁵ Target module: SCI4 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
28	MSTPB28	Serial Communication Interface 3 Module Stop* ⁵ Target module: SCI3 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
29	MSTPB29	Serial Communication Interface 2 Module Stop* ⁵ Target module: SCI2 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
30	MSTPB30	Serial Communication Interface 1 Module Stop* ⁵ Target module: SCI1 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
31	MSTPB31	Serial Communication Interface 0 Module Stop* ⁵ Target module: SCI0 0: Cancel the module-stop state 1: Enter the module-stop state	R/W

Note: S-TYPE-4, P-TYPE-4

- Note 1. The MSTPB4 bit should be written after the I3C clock (I3CCLK) is stable. To enter Software Standby mode, the WFI instruction should be executed after 2 or more cycles of I3CCLK following the change of MSTPB4.
- Note 2. The MSTPB11 bit should be written after the USB clock (USBCLK) is stable. To enter Software Standby mode, the WFI instruction should be executed after 2 or more cycles of USBCLK following the change of MSTPB11.
- Note 3. The MSTPB16 and MSTPB17 bits should be written after the Octal-SPI clock (OCTACLK) is stable. To enter Software Standby mode, the WFI instruction should be executed after 2 or more cycles of OCTACLK following the change of MSTPB16 and MSTPB17. When changing the value of this bit, only execute subsequent instructions after reading this bit to check that the value was updated.
- Note 4. The MSTPB18 and MSTPB19 bits should be written after the SPI clock (SPICLK) is stable. To enter Software Standby mode, the WFI instruction should be executed after 2 or more cycles of SPICLK following the change of MSTPB18 and MSTPB19.
- Note 5. The MSTPBi bits should be written after the SCI clock (SCICLK) is stable. To enter Software Standby mode, the WFI instruction should be executed after 2 or more cycles of SCICLK following the change of MSTPBi (i = 22 to 31).

11.2.8 MSTPCRC : Module Stop Control Register C

Base address: MSTP = 0x4020_3000
MSTP_NS = 0x5020_3000

Offset address: 0x08

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	MSTP C31	MSTP C30	MSTP C29	MSTP C28	MSTP C27	MSTP C26	—	—	—	—	—	—	—	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	MSTP C14	MSTP C13	MSTP C12	MSTP C11	—	—	—	—	—	—	—	—	—	MSTP C1	MSTP C0
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

The MSTPC26 and MSTPC27 bits should be written after the CANFD clock (CANFDCLK) is stable. To enter Software Standby mode, the WFI instruction should be executed after 2 or more cycles of CANFDCLK following the change of MSTPC26 and MSTPC27.

The MSTPC29 bit should be written after the EtherCAT clock (ESCCLK) is stable. To enter Software Standby mode, the WFI instruction should be executed after 2 or more cycles of ESCCLK following the change of MSTPC29.

Bit	Symbol	Function	R/W
0	MSTPC0	Clock Frequency Accuracy Measurement Circuit Module Stop*1 Target module: CAC 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
1	MSTPC1	Cyclic Redundancy Check Calculator Module Stop Target module: CRC 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
10:2	—	These bits are read as 1. The write value should be 1.	R/W
11	MSTPC11	Secure Digital Host IF 1 Module Stop Target module: SDHI1 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
12	MSTPC12	Secure Digital Host IF 0 Module Stop Target module: SDHI0 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
13	MSTPC13	Data Operation Circuit Module Stop Target module: DOC 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
14	MSTPC14	Event Link Controller Module Stop Target module: ELC 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
25:15	—	These bits are read as 1. The write value should be 1.	R/W

The MSTPC26 and MSTPC27 bits should be written after the CANFD clock (CANFDCLK) is stable. To enter Software Standby mode, the WFI instruction should be executed after 2 or more cycles of CANFDCLK following the change of MSTPC26 and MSTPC27. The MSTPC29 bit should be written after the EtherCAT clock (ESCCLK) is stable. To enter Software Standby mode, the WFI instruction should be executed after 2 or more cycles of ESCCLK following the change of MSTPC29.

Bit	Symbol	Function	R/W
26	MSTPC26	Controller Area Network with Flexible Data-Rate 1 Module Stop The MSTPC26 and MSTPC27 bits should be written after the CANFD clock (CANFDCLK) is stable. To enter Software Standby mode, the WFI instruction should be executed after 2 or more cycles of CANFDCLK following the change of MSTPC26 and MSTPC27. Target module: CANFD1 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
27	MSTPC27	Controller Area Network with Flexible Data-Rate 0 Module Stop The MSTPC26 and MSTPC27 bits should be written after the CANFD clock (CANFDCLK) is stable. To enter Software Standby mode, the WFI instruction should be executed after 2 or more cycles of CANFDCLK following the change of MSTPC26 and MSTPC27. Target module: CANFD0 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
28	MSTPC28	Ether-PHY clock Module Stop ^{*2} Target module: ETHPHYCLK 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
29	MSTPC29	EtherCAT Slave Controller Module Stop The MSTPC29 bit should be written after the EtherCAT clock (ESCCLK) is stable. To enter Software Standby mode, the WFI instruction should be executed after 2 or more cycles of ESCCLK following the change of MSTPC29. Target module: ESC 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
30	MSTPC30	Layer 3 Ethernet Switch Module Module Stop ^{*3} Target module: ESWM 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
31	MSTPC31	Renesas Secure IP Module Stop Target module: RSIP-E50D 0: Cancel the module-stop state 1: Enter the module-stop state	R/W

Note: S-TYPE-4, P-TYPE-4

Note 1. The MSTPC0 bit should be written after the CAC clock is stable. To enter Software Standby mode, the WFI instruction should be executed after 2 or more cycles of the CAC clock following the change of MSTPC0.

Note 2. The MSTPC28 bit should be written after the Ether-PHY clock (ETHPHYCLK) is stable. To enter Software Standby mode, the WFI instruction should be executed after 2 or more cycles of ETHPHYCLK following the change of MSTPC28.

Note 3. The MSTPC30 bit should be written after the EtherSW clock (ESWCLK) is stable. To enter Software Standby mode, the WFI instruction should be executed after 2 or more cycles of ESWCLK following the change of MSTPC30.

11.2.9 MSTPCRD : Module Stop Control Register D

Base address: MSTP = 0x4020_3000
MSTP_NS = 0x5020_3000

Offset address: 0x0C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	MSTP D28	MSTP D27	MSTP D26	MSTP D25	—	—	MSTP D22	MSTP D21	MSTP D20	MSTP D19	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	MSTP D14	MSTP D13	MSTP D12	MSTP D11	—	MSTP D9	MSTP D8	—	MSTP D6	MSTP D5	MSTP D4	—	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
3:0	—	These bits are read as 1. The write value should be 1.	R/W
4	MSTPD4	Low power Asynchronous General Purpose Timer 1 Module Stop* ¹ Target module: AGT1 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
5	MSTPD5	Low power Asynchronous General Purpose Timer 0 Module Stop* ¹ Target module: AGT0 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
6	MSTPD6	PWM Delay Generation Circuit Module Stop Target module: PDG 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
7	—	This bit is read as 1. The write value should be 1.	R/W
8	MSTPD8	Delta-Sigma Modulator Interface 1 Module Stop* ² Target module: DSMIF1 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
9	MSTPD9	Delta-Sigma Modulator Interface 0 Module Stop* ² Target module: DSMIF0 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
10	—	This bit is read as 1. The write value should be 1.	R/W
11	MSTPD11	Port Output Enable for GPT Group D Module Stop Target module: POEG Group D 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
12	MSTPD12	Port Output Enable for GPT Group C Module Stop Target module: POEG Group C 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
13	MSTPD13	Port Output Enable for GPT Group B Module Stop Target module: POEG Group B 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
14	MSTPD14	Port Output Enable for GPT Group A Module Stop Target module: POEG Group A 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
18:15	—	These bits are read as 1. The write value should be 1.	R/W
19	MSTPD19	12-bit D/A Converter 1 Module Stop Target module: DAC12_1 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
20	MSTPD20	12-bit D/A Converter 0 Module Stop Target module: DAC12_0 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
21	MSTPD21	16-Bit A/D Converter Module Stop* ³ Target module: ADC16H 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
22	MSTPD22	Temperature Sensor Module Stop Target module: TSN 0: Cancel the module-stop state 1: Enter the module-stop state	R/W

Bit	Symbol	Function	R/W
24:23	—	These bits are read as 1. The write value should be 1.	R/W
25	MSTPD25	High-Speed Analog Comparator 3 Module Stop Target module: ACMPHS3 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
26	MSTPD26	High-Speed Analog Comparator 2 Module Stop Target module: ACMPHS2 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
27	MSTPD27	High-Speed Analog Comparator 1 Module Stop Target module: ACMPHS1 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
28	MSTPD28	High-Speed Analog Comparator 0 Module Stop Target module: ACMPHS0 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
31:29	—	These bits are read as 1. The write value should be 1.	R/W

Note: S-TYPE-4, P-TYPE-4

Note 1. When the count source is sub-clock oscillator or LOCO, AGTn counting doesn't stop even if MSTPD_i is set to 1. If the count source is the sub-clock oscillator or LOCO, this bit must be set to 1 except when accessing the AGTn registers. (n = 1 to 0) (i = 4 to 5)

Note 2. The MSTPD8 and MSTPD9 bits should be written after the DSMIF clock (DSMIFCLK) is stable. To enter Software Standby mode, the WFI instruction should be executed after 2 or more cycles of DSMIFCLK following the change of MSTPD8 and MSTPD9.

Note 3. The MSTPD21 bit should be written after the ADC clock (ADCCLK) is stable. To enter Software Standby mode, the WFI instruction should be executed after 2 or more cycles of ADCCLK following the change of MSTPD21.

11.2.10 MSTPCRE : Module Stop Control Register E

Base address: MSTP = 0x4020_3000
MSTP_NS = 0x5020_3000

Offset address: 0x10

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	MSTP E31	MSTP E30	MSTP E29	MSTP E28	MSTP E27	—	—	—	—	—	MSTP E21	MSTP E20	MSTP E19	MSTP E18	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	MSTP E9	MSTP E8	—	—	—	—	—	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
7:0	—	These bits are read as 1. The write value should be 1.	R/W
8	MSTPE8	Ultra-Low Power Timer 1 Module Stop*1 Target module: ULPT1 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
9	MSTPE9	Ultra-Low Power Timer 0 Module Stop*1 Target module: ULPT0 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
17:10	—	These bits are read as 1. The write value should be 1.	R/W
18	MSTPE18	General PWM Timer 13 Module Stop*2 Target module: GPT13 0: Cancel the module-stop state 1: Enter the module-stop state	R/W

Bit	Symbol	Function	R/W
19	MSTPE19	General PWM Timer 12 Module Stop* ² Target module: GPT12 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
20	MSTPE20	General PWM Timer 11 Module Stop* ² Target module: GPT11 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
21	MSTPE21	General PWM Timer 10 Module Stop* ² Target module: GPT10 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
26:22	—	These bits are read as 1. The write value should be 1.	R/W
27	MSTPE27	General PWM Timer 4-9 Module Stop* ² Target module: GPT4-9 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
28	MSTPE28	General PWM Timer 3 Module Stop* ² Target module: GPT3 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
29	MSTPE29	General PWM Timer 2 Module Stop* ² Target module: GPT2 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
30	MSTPE30	General PWM Timer 1 Module Stop* ² Target module: GPT1 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
31	MSTPE31	General PWM Timer 0 Module Stop* ² Target module: GPT0 0: Cancel the module-stop state 1: Enter the module-stop state	R/W

Note: S-TYPE-4, P-TYPE-4

Note 1. When the count source is sub-clock oscillator or LOCO, ULPTn counting does not stop even if MSTPEi is set to 1. If the count source is the sub-clock oscillator or LOCO, this bit must be set to 1 except when accessing the ULPTn registers. (n = 1, 0) (i = 8, 9)

Note 2. The MSTPEi bits should be written after the GPT clock (GPTCLK) is stable. To enter Software Standby mode, the WFI instruction should be executed after 2 or more cycles of GPTCLK following the change of MSTPEi (i = 18 to 21, 27 to 31).

11.2.11 OPCCR : Operating Power Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x0A0

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	OPCM TSF	—	—	OPCM[1:0]	

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
1:0	OPCM[1:0]	Operating Power Control Mode Select 0 0: High-speed mode 0 1: Prohibited 1 0: Prohibited 1 1: Prohibited Settings other than above are prohibited.	R/W

Bit	Symbol	Function	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
4	OPCMTSF	Operating Power Control Mode Transition Status Flag - Read 0: Transition completed 1: During transition - Write - The write value should be 0.	R
7:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

OPCCR is used to reduce power consumption in Normal mode and Processor low power mode. Do not change from the initial value.

11.2.12 PGSCR: Power Gating Shift Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x104

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	PGS
Value after reset:	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
0	PGS	Power gating shift timing Set the power gating start timing for each domain. Set this bit to 0 before transitioning to Software Standby mode or Deep Software Standby mode. 0: Power gating control for each domain is started partial at the same time 1: Power gating control for each domain is started serially	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

11.2.13 PSSTCRn : Power Switch Control Start Time Control Register n (n = 2 to 5)

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x210 + 0x2 × n (n = 2 to 5)

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	PSST												
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
12:0	PSST	Power switch control start time Set the power switch control start time when PGSCR.PGS = 0. Set 0x00B0 to this bit. Other setting is prohibited.	R/W
15:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

11.2.14 PDCTRESWM : ESWM Power Domain Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x118

Bit position:	7	6	5	4	3	2	1	0
Bit field:	PDPG SF	PDCS F	—	—	—	—	—	PDDE
Value after reset:	1	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
0	PDDE	Power control enable 0: Power on the target domain 1: Power off the target domain	R/W
5:1	—	These bits are read as 0. The write value should be 0.	R/W
6	PDCSF	Power control status flag 0: Power gating control is not executed (idle) 1: Power gating control is in progress	R
7	PDPGSF	Power gating status flag 0: Target domain is power on (not gating) 1: Target domain is power off (during Gating)	R

Note: S-TYPE-3, P-TYPE-2

The PDCTRESWM register controls power gating for ESWM power domain.

For restrictions on this register, not only in this section, see also [section 11.5.1. Power Gating Control Using Power Domain Control Register](#).

PDDE bit (Power control enable)

The PDDE bit controls the power gating of the target domain. The PDDE bit should be set from 0 to 1, after confirmed that the PDCSF = 0 and PDPGSF = 0. On the other hand, the PDDE bit should be set from 1 to 0, after confirmed that the PDCSF = 0 and PDPGSF = 1.

PDCSF flag (Power control status flag)

The PDCSF flag indicates the power control status of the target domain.

PDPGSF flag (Power gating status flag)

The PDPGSF flag indicates whether the target domain is power gated.

11.2.15 PDRAMSCR0 : SRAM Power Domain Standby Control Register 0

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x140

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	RKEE P12	RKEE P11	RKEE P10	RKEE P9	RKEE P8	RKEE P7	RKEE P6	RKEE P5	RKEE P4	RKEE P3	RKEE P2	RKEE P1	RKEE P0
Value after reset:	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
12:0	RKEEP[12:0]	RAM Retention 0: When entering the Software Standby mode, the contents of the target RAM are not kept. 1: When entering the Software Standby mode, the contents of the target RAM are kept.	R/W
14:13	—	These bits are read as 1. The write value should be 1.	R/W

Bit	Symbol	Function	R/W
15	—	This bit is read as 0. The write value should be 0.	R/W

Note: S-TYPE-4, P-TYPE-2

The PDRAMSCR0 register controls the target memory in Software Standby mode.

RKEEPn bit (RAM Retention)

The RKEEPn (n = 0 to 12) bit controls whether to retain the contents of the target memory in software standby mode.

Table 11.5 shows the relationship between each bit and the target memory.

In addition, there are the following constraints regarding the setting of each bit.

RKEEP0 to RKEEP3

0000: When entering the Software Standby mode, the contents of the SRAM0 are not kept.

1111: When entering the Software Standby mode, the contents of the SRAM0 are kept.

Other than the above: Not supported

RKEEP4 to RKEEP7

0000: When entering the Software Standby mode, the contents of the SRAM1 are not kept.

1111: When entering the Software Standby mode, the contents of the SRAM1 are kept.

Other than the above: Not supported

RKEEP8 to RKEEP11

0000: When entering the Software Standby mode, the contents of the SRAM2 are not kept.

1111: When entering the Software Standby mode, the contents of the SRAM2 are kept.

Other than the above: Not supported

Table 11.5 Relationship between each bit and the target memory (1 of 2)

PDRAM register bit	Target memory
PDRAM.b0	SRAM0: 0x2200_0000 to 0x2201_FFFF (Secure alias) 0x3200_0000 to 0x3201_FFFF (Non-secure alias) SRAM (CPU1 TCM)*1: 0x2A00_0000 to 0x2A01_FFFF (Secure alias) 0x3A00_0000 to 0x3A01_FFFF (Non-secure alias)
PDRAM.b1	SRAM0: 0x2202_0000 to 0x2203_FFFF (Secure alias) 0x3202_0000 to 0x3203_FFFF (Non-secure alias)
PDRAM.b2	SRAM0: 0x2204_0000 to 0x2205_FFFF (Secure alias) 0x3204_0000 to 0x3205_FFFF (Non-secure alias)
PDRAM.b3	SRAM0: 0x2206_0000 to 0x2207_FFFF (Secure alias) 0x3206_0000 to 0x3207_FFFF (Non-secure alias)
PDRAM.b4	SRAM1: 0x2208_0000 to 0x2209_FFFF (Secure alias) 0x3208_0000 to 0x3209_FFFF (Non-secure alias)
PDRAM.b5	SRAM1: 0x220A_0000 to 0x220B_FFFF (Secure alias) 0x320A_0000 to 0x320B_FFFF (Non-secure alias)
PDRAM.b6	SRAM1: 0x220C_0000 to 0x220D_FFFF (Secure alias) 0x320C_0000 to 0x320D_FFFF (Non-secure alias)
PDRAM.b7	SRAM1: 0x220E_0000 to 0x220F_FFFF (Secure alias) 0x320E_0000 to 0x320F_FFFF (Non-secure alias)

Table 11.5 Relationship between each bit and the target memory (2 of 2)

PDRAM register bit	Target memory
PDRAM.b8	SRAM2: 0x2210_0000 to 0x2211_FFFF (Secure alias) 0x3210_0000 to 0x3211_FFFF (Non-secure alias)
PDRAM.b9	SRAM2: 0x2212_0000 to 0x2213_FFFF (Secure alias) 0x3212_0000 to 0x3213_FFFF (Non-secure alias)
PDRAM.b10	SRAM2: 0x2214_0000 to 0x2215_FFFF (Secure alias) 0x3214_0000 to 0x3215_FFFF (Non-secure alias)
PDRAM.b11	SRAM2: 0x2216_0000 to 0x2217_FFFF (Secure alias) 0x3216_0000 to 0x3217_FFFF (Non-secure alias)
PDRAM.b12	SRAM3: 0x2218_0000 to 0x2219_FFFF (Secure alias) 0x3218_0000 to 0x3219_FFFF (Non-secure alias)

Note 1. SRAM (CPU1 TCM) is supported only for single core product.

11.2.16 PDRAMSCR1 : SRAM Power Domain Standby Control Register 1

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x142

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	RKEE P1	RKEE P0

Value after reset: 0 0 0 0 0 0 0 1 1

Bit	Symbol	Function	R/W
0	RKEEP0	RAM Retention 0: When entering the CPU0 Deep Sleep and Software Standby mode, the contents of the target RAM are not kept. 1: When entering the CPU0 Deep Sleep and Software Standby mode, the contents of the target RAM are kept.	R/W
1	RKEEP1 ^{*1}	RAM Retention 0: When entering the Software Standby mode, the contents of the target RAM are not kept. 1: When entering the Software Standby mode, the contents of the target RAM are kept.	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-4, P-TYPE-2

Note 1. RKEEP1 is valid only for a dual-core product.

The PDRAMSCR1 register controls the target memory in CPU0 Deep Sleep and Software Standby mode.

RKEEP0 bit (RAM Retention)

The RKEEP0 bit controls whether to retain the contents of the target memory in CPU0 Deep Sleep and software standby mode.

RKEEP1 bit (RAM Retention)

The RKEEP1 bit controls whether to retain the contents of the target memory in software standby mode.

Table 11.6 shows the relationship between each bit and the target memory.

Table 11.6 Relationship between each bit and the target memory

PDRAM register bit	Target memory
PDRAM.b0	CPU0 TCM
PDRAM.b1 ^{*1}	CPU1 TCM

Note 1. Available only for a dual-core product.

11.2.17 SBYCR : Standby Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x00C

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	OPE	—	—	—	—	—	—
Value after reset:	0	1	0	0	0	0	0	0

Bit	Symbol	Function	R/W
5:0	—	These bits are read as 0. The write value should be 0.	R/W
6	OPE	Output Port Enable 0: In Software Standby mode or Deep Software Standby mode, Set the address bus and other bus control signal to the high-impedance state. 1: In Software Standby mode or Deep Software Standby mode, Address bus and other bus control signal retain the output state.	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

The SBYCR register controls the bus signal output in Software Standby mode and Deep Software Standby mode.

OPE bit (Output Port Enable)

The OPE bit specifies whether to set to the high-impedance state or to retain the output of the address bus and bus control signals (For SRAM: [A00 to A23, CS0 to CS7, RD, WR0 to WR3, WR, BC0 to BC3 and ALE], and for SDRAM: [A00 to A16, DQM0 to DQM3, SDCS, RAS, CAS, WE and CKE]) in Software Standby mode or Deep Software Standby mode.

11.2.18 SSCR1 : Software Standby Control Register 1

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xA98

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	SS2LP[1:0]	—	—	SS2FR
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SS2FR	Software Standby Fast Return 0: When returning from Software Standby mode, fast return function is disabled. 1: When returning from Software Standby mode, fast return function is enabled.	R/W
1	—	This bit is read as 0. The write value should be 0.	R/W
3:2	SS2LP[1:0]	Software Standby low power select 0 0: SS2LP_0 (default mode) 0 1: SS2LP_1 (Low power setting 1) Settings other than above are prohibited.	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

SS2FR bit (Software Standby Fast Return)

When the SS2FR bit is 1, the recovery time from Software Standby mode is shorter. The setting of this bit is only valid when the SS2LP bit is 01b.

See [section 60, Electrical Characteristics](#) for details on the recovery time.

Note: It is recommended to set this bit to 1. Otherwise, it just increases the recovery time.

SS2LP[1:0] bits (Software Standby low power select)

When the SS2LP[1:0] bits are 01b, the power consumption of Software Standby mode is smaller but the recovery time from Software Standby mode is longer. See [section 60, Electrical Characteristics](#) for details on the power consumption and the recovery time.

11.2.19 LPSCR : Low Power State Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xA90

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	LPMD[3:0]			
Value after reset:	0	0	0	0	0	1	0	0

Bit	Symbol	Function	R/W
3:0	LPMD[3:0]	Low power mode setting 0x0: System Active 0x4: Reserved*1 0x5: Software Standby mode 0x8: Deep Software Standby mode 1 0x9: Deep Software Standby mode 2 0xA: Deep Software Standby mode 3 Settings other than above are prohibited.	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

Note 1. Setting prohibited when transitioning to low power mode. Change the setting to a value other than 0x4 before transitioning to low power mode.

The LPSCR register controls the transition destination of low power mode.

LPSCR is not initialized by the internal reset signal (Deep Software Standby reset signal) that is the source to cancel the Deep Software Standby mode. For details, see [section 6, Resets](#).

LPMD[3:0] bits (Low power mode setting)

The LPMD[3:0] bits indicate the destination of low power mode. Actual low power mode transition is executed by WFI instruction in CPU.

The transition of Software Standby mode and Deep Software Standby mode does not occur unless the WFI instruction of each CPU is executed.

When the low power mode is canceled by an interrupt, The LPMD bit remains the value before transition. Writing 0 to this bit clears the bit itself.

While MENTRYR.MENTRY bit is 1, the setting to Software Standby and Deep software Standby mode is ineffective. Even if LPMD[3:0] is any of 0x5, 0x8 to 0xA, the MCU is keep to System active state on execution of a WFI instruction. See [Table 11.8](#) for the detail.

While OFS0.IWDTSTPCTL bit is 0 (counting continues), the setting to Deep software Standby 2 and 3 mode is ineffective. Even if LPMD is any of 0x9 or 0xA, the MCU enters Deep Software Standby 1 mode on execution of a WFI instruction.

While IWDTCSTPR.SLCSTP is 0 (counting continues), the setting to Deep software Standby 2 and 3 mode is ineffective. Even if LPMD is any of 0x9 or 0xA, the MCU enters Deep Software Standby 1 mode on execution of a WFI instruction.

See [Table 11.8](#) for the detail.

While the voltage monitor n ($n = 1, 2$) reset is enabled ($PVDnCR0.RI = 1$), the setting to Deep software Standby mode 2, 3 is ineffective. Even if $LPMD$ is any of $0x9$ or $0xA$, the MCU enters Deep Software Standby 1 mode on execution of a WFI instruction. See [Table 11.8](#) for the detail.

When at least $PVDnCR0.RE$ ($n = 4, 5$) bit is 1 and all $PVDnCR0.RI$ ($n = 1, 2$) bit are 0 and $OFS0.IWDTSTPCTL$ is 1 and $IWDTCSSTPR.SLCSTP$ is 1, the setting to Deep software Standby mode 3 is ineffective. Even if $LPMD$ is $0xA$, the MCU enters Deep Software Standby 2 mode on execution of a WFI instruction. See [Table 11.8](#) for the detail.

11.2.20 DPSBYCR : Deep Software Standby Control Register

Base address: SYSC = $0x4001_E000$
SYSC_NS = $0x5001_E000$

Offset address: $0xA00$

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	IOKEEP P	—	—	DCSSMODE [1:0]	—	—	—
Value after reset:	0	0	0	1	0	1	0	0

Bit	Symbol	Function	R/W
1:0	—	These bits are read as 0. The write value should be 0.	R/W
3:2	DCSSMODE[1:0]	DCDC soft start mode Set the wake-up time from the Deep Software Standby mode with DCDC. 0 0: Setting prohibited 0 1: 128 μ s 1 0: 256 μ s 1 1: 512 μ s	R/W
5:4	—	These bits are read as 0. The write value should be 0.	R/W
6	IOKEEP	I/O Port Retention 0: When the Deep Software Standby mode is canceled, the I/O ports and D/A output are in the reset state 1: When the Deep Software Standby mode is canceled, the I/O ports and D/A output are in the same state as in the Deep Software Standby mode	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

The DPSBYCR register controls the Deep Software Standby mode.

DPSBYCR is not initialized by the internal reset signal (Deep Software Standby reset signal) that is the source to cancel the Deep Software Standby mode. For details, see [section 6, Resets](#).

DCSSMODE[1:0] bits (DCDC soft start mode)

The DCSSMODE[1:0] bits specify whether to shorten the time required to recover from the Deep Software Standby mode. For details, see [section 60, Electrical Characteristics](#).

IOKEEP bit (I/O Port Retention)

In Deep Software Standby mode, I/O ports and D/A output keep the same states as in the Software Standby mode. The IOKEEP bit specifies whether to reset the state of the I/O ports and D/A output or not when the Deep Software Standby mode is canceled.

11.2.21 DPSIER0 : Deep Software Standby Interrupt Enable Register 0

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xA08

Bit position:	7	6	5	4	3	2	1	0
Bit field:	DIRQ7 E	DIRQ6 E	DIRQ5 E	DIRQ4 E	DIRQ3 E	DIRQ2 E	DIRQ1 E	DIRQ0 E
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DIRQ0E	IRQ0-DS Pin Enable 0: Cancelling Deep Software Standby mode is disabled 1: Cancelling Deep Software Standby mode is enabled	R/W
1	DIRQ1E	IRQ1-DS Pin Enable 0: Cancelling Deep Software Standby mode is disabled 1: Cancelling Deep Software Standby mode is enabled	R/W
2	DIRQ2E	IRQ2-DS Pin Enable 0: Cancelling Deep Software Standby mode is disabled 1: Cancelling Deep Software Standby mode is enabled	R/W
3	DIRQ3E	IRQ3-DS Pin Enable 0: Cancelling Deep Software Standby mode is disabled 1: Cancelling Deep Software Standby mode is enabled	R/W
4	DIRQ4E	IRQ4-DS Pin Enable 0: Cancelling Deep Software Standby mode is disabled 1: Cancelling Deep Software Standby mode is enabled	R/W
5	DIRQ5E	IRQ5-DS Pin Enable 0: Cancelling Deep Software Standby mode is disabled 1: Cancelling Deep Software Standby mode is enabled	R/W
6	DIRQ6E	IRQ6-DS Pin Enable 0: Cancelling Deep Software Standby mode is disabled 1: Cancelling Deep Software Standby mode is enabled	R/W
7	DIRQ7E	IRQ7-DS Pin Enable 0: Cancelling Deep Software Standby mode is disabled 1: Cancelling Deep Software Standby mode is enabled	R/W

Note: S-TYPE-4, P-TYPE-2

DPSIER0 is not initialized by the internal reset signal (Deep Software Standby reset signal) that is the source to cancel the Deep Software Standby mode. For details, see [section 6, Resets](#).

After the setting of DPSIER0 is modified, an edge may be internally generated depending on the state of the pin, resulting in DPSIFR0 being set to 1. Therefore, DPSIFR0 should be cleared to 0 before entering Deep Software Standby mode.

See [Table 11.4](#) to know the relation between this register and interrupt source for canceling Deep Software standby mode.

11.2.22 DPSIER1 : Deep Software Standby Interrupt Enable Register 1

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xA0C

Bit position:	7	6	5	4	3	2	1	0
Bit field:	DIRQ1 5E	DIRQ1 4E	DIRQ1 3E	DIRQ1 2E	DIRQ1 1E	DIRQ1 0E	DIRQ9 E	DIRQ8 E
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DIRQ8E	IRQ8-DS Pin Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W
1	DIRQ9E	IRQ9-DS Pin Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W
2	DIRQ10E	IRQ10-DS Pin Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W
3	DIRQ11E	IRQ11-DS Pin Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W
4	DIRQ12E	IRQ12-DS Pin Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W
5	DIRQ13E	IRQ13-DS Pin Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W
6	DIRQ14E	IRQ14-DS Pin Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W
7	DIRQ15E	IRQ15-DS Pin Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W

Note: S-TYPE-4, P-TYPE-2

DPSIER1 is not initialized by the internal reset signal (Deep Software Standby reset signal) that is the source to cancel the Deep Software Standby mode. For details, see [section 6, Resets](#).

After the setting of DPSIER1 is modified, an edge may be internally generated depending on the state of the pin, resulting in DPSIFR1 being set to 1. Therefore, DPSIFR1 should be cleared to 0 before entering Deep Software Standby mode.

See [Table 11.4](#) to know the relation between this register and interrupt source for canceling Deep Software standby mode.

11.2.23 DPSIER2 : Deep Software Standby Interrupt Enable Register 2

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xA10

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	DNMI E	DRTC AIE	DRTC IE	DPVD 2IE	DPVD 1IE

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	DPVD1IE	PVD1 Deep Software Standby Cancel Signal Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W
1	DPVD2IE	PVD2 Deep Software Standby Cancel Signal Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W
2	DRTCIE	RTC Interval interrupt Deep Software Standby Cancel Signal Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W
3	DRTCAIE	RTC Alarm interrupt Deep Software Standby Cancel Signal Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W

Bit	Symbol	Function	R/W
4	DNMIE	NMI Pin Deep Software Standby Cancel Signal Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W ^{*1}
7:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-4, P-TYPE-2

Note 1. 1 can be written only once. Once 1 is written to this bit, subsequent write accesses are disabled.

DPSIER2 is not initialized by the internal reset signal (Deep Software Standby reset signal) that is the source to cancel the Deep Software Standby mode. For details, see [section 6, Resets](#).

After the setting of DPSIER2 is modified, an edge may be internally generated depending on the state of the pin, resulting in DPSIFR2 being set to 1. Therefore, DPSIFR2 should be cleared to 0 before entering Deep Software Standby mode.

See [Table 11.4](#) to know the relation between this register and interrupt source for canceling Deep Software standby mode.

11.2.24 DPSIER3 : Deep Software Standby Interrupt Enable Register 3

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xA14

Bit position:	7	6	5	4	3	2	1	0
Bit field:	DVBATTADIE	DSOSTDIE	DIWD TIE	—	DULPT1IE	DULPT0IE	—	DUSBFSIE
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DUSBFSIE	USBFS Suspend/Resume Deep Software Standby Cancel Signal Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W
1	—	This bit is read as 0. The write value should be 0.	R/W
2	DULPT0IE	ULPT0 Overflow Deep Software Standby Cancel Signal Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W
3	DULPT1IE	ULPT1 Overflow Deep Software Standby Cancel Signal Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W
4	—	This bit is read as 0. The write value should be 0.	R/W
5	DIWD TIE	IWDT Overflow Deep Software Standby Cancel Signal Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W
6	DSOSTDIE	Sub-clock Oscillation stop detection Deep Software Standby Cancel Signal Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W
7	DVBATTADIE	VBATT Tamper Detection Deep Software Standby Cancel Signal Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W

Note: S-TYPE-4, P-TYPE-2

DPSIER3 is not initialized by the internal reset signal (Deep Software Standby reset signal) that is the source to cancel the Deep Software Standby mode. For details, see [section 6, Resets](#).

After the setting of DPSIER3 is modified, an edge may be internally generated depending on the state of the pin, resulting in DPSIFR3 being set to 1. Therefore, DPSIFR3 should be cleared to 0 before entering Deep Software Standby mode.

See [Table 11.4](#) to know the relation between this register and interrupt source for canceling Deep Software standby mode.

11.2.25 DPSIER4 : Deep Software Standby Interrupt Enable Register 4

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xB40

Bit position:	7	6	5	4	3	2	1	0
Bit field:	DIRQ2 3E	DIRQ2 2E	DIRQ2 1E	DIRQ2 0E	DIRQ1 9E	DIRQ1 8E	DIRQ1 7E	DIRQ1 6E

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	DIRQ16E	IRQ16-DS Pin Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W
1	DIRQ17E	IRQ17-DS Pin Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W
2	DIRQ18E	IRQ18-DS Pin Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W
3	DIRQ19E	IRQ19-DS Pin Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W
4	DIRQ20E	IRQ20-DS Pin Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W
5	DIRQ21E	IRQ21-DS Pin Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W
6	DIRQ22E	IRQ22-DS Pin Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W
7	DIRQ23E	IRQ23-DS Pin Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W

Note: S-TYPE-4, P-TYPE-2

DPSIER4 is not initialized by the internal reset signal (Deep Software Standby reset signal) that is the source to cancel the Deep Software Standby mode. For details, see [section 6, Resets](#).

After the setting of DPSIER4 is modified, an edge may be internally generated depending on the state of the pin, resulting in DPSIFR4 being set to 1. Therefore, DPSIFR4 should be cleared to 0 before entering Deep Software Standby mode.

See [Table 11.4](#) to know the relation between this register and interrupt source for canceling Deep Software standby mode.

11.2.26 DPSIER5 : Deep Software Standby Interrupt Enable Register 5

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xB44

Bit position:	7	6	5	4	3	2	1	0
Bit field:	DIRQ3 1E	DIRQ3 0E	DIRQ2 9E	DIRQ2 8E	DIRQ2 7E	DIRQ2 6E	DIRQ2 5E	DIRQ2 4E

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	DIRQ24E	IRQ24-DS Pin Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W
1	DIRQ25E	IRQ25-DS Pin Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W
2	DIRQ26E	IRQ26-DS Pin Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W
3	DIRQ27E	IRQ27-DS Pin Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W
4	DIRQ28E	IRQ28-DS Pin Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W
5	DIRQ29E	IRQ29-DS Pin Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W
6	DIRQ30E	IRQ30-DS Pin Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W
7	DIRQ31E	IRQ31-DS Pin Enable 0: Canceling Deep Software Standby mode is disabled 1: Canceling Deep Software Standby mode is enabled	R/W

Note: S-TYPE-4, P-TYPE-2

DPSIER5 is not initialized by the internal reset signal (Deep Software Standby reset signal) that is the source to cancel the Deep Software Standby mode. For details, see [section 6, Resets](#).

After the setting of DPSIER5 is modified, an edge may be internally generated depending on the state of the pin, resulting in DPSIFR5 being set to 1. Therefore, DPSIFR5 should be cleared to 0 before entering Deep Software Standby mode.

See [Table 11.4](#) to know the relation between this register and interrupt source for canceling Deep Software standby mode.

11.2.27 DPSIFR0 : Deep Software Standby Interrupt Flag Register 0

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xA18

Bit position:	7	6	5	4	3	2	1	0
Bit field:	DIRQ7 F	DIRQ6 F	DIRQ5 F	DIRQ4 F	DIRQ3 F	DIRQ2 F	DIRQ1 F	DIRQ0 F

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	DIRQ0F	IRQ0-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ¹
1	DIRQ1F	IRQ1-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ¹
2	DIRQ2F	IRQ2-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ¹
3	DIRQ3F	IRQ3-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ¹

Bit	Symbol	Function	R/W
4	DIRQ4F	IRQ4-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}
5	DIRQ5F	IRQ5-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}
6	DIRQ6F	IRQ6-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}
7	DIRQ7F	IRQ7-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}

Note: S-TYPE-4, P-TYPE-2

Note 1. Only 0 can be written to clear flag. The flag must be cleared by writing 0 after 1 is read.

Each flag is set to 1 when a cancel request specified by DPSIEGR0 is generated.

Each flag may be set to 1 when a cancel request is generated in any mode (not even Deep Software Standby mode) or when the setting of DPSIER0 is modified. Therefore, a transition to Deep Software Standby mode should be made after DPSIFR0 is cleared to 0x00.

To clear DPSIFR0 to 0x00 after modifying DPSIER0, wait for at least six PCLKB cycles, read DPSIFR0, and then write 0 to DPSIFR0. Six or more PCLKB cycles can be secured, for example, by reading DPSIER0.

DPSIFR0 is not initialized by the internal reset signal (Deep Software Standby reset signal) that is the source to cancel the Deep Software Standby mode. For details, see [section 6, Resets](#).

DIRQnF flag (IRQn-DS Pin Deep Software Standby Cancel Flag) (n = 0 to 7)

The DIRQnF flag indicates that a cancel request by the IRQn-DS pin has been generated.

[Setting condition]

A cancel request by the IRQn-DS pin specified by DPSIEGR0 is generated.

[Clearing condition]

Writing 0 to each flag after 1 is read.

11.2.28 DPSIFR1 : Deep Software Standby Interrupt Flag Register 1

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xA1C

Bit position:	7	6	5	4	3	2	1	0
Bit field:	DIRQ1 5F	DIRQ1 4F	DIRQ1 3F	DIRQ1 2F	DIRQ1 1F	DIRQ1 0F	DIRQ9 F	DIRQ8 F

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	DIRQ8F	IRQ8-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}
1	DIRQ9F	IRQ9-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}
2	DIRQ10F	IRQ10-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}

Bit	Symbol	Function	R/W
3	DIRQ11F	IRQ11-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ¹
4	DIRQ12F	IRQ12-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ¹
5	DIRQ13F	IRQ13-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ¹
6	DIRQ14F	IRQ14-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ¹
7	DIRQ15F	IRQ15-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ¹

Note: S-TYPE-4, P-TYPE-2

Note 1. Only 0 can be written to clear flag. The flag must be cleared by writing 0 after 1 is read.

Each flag is set to 1 when a cancel request specified by DPSIEGR1 is generated.

Each flag may be set to 1 when a cancel request is generated in any mode (not even Deep Software Standby mode) or when the setting of DPSIER1 is modified. Therefore, a transition to Deep Software Standby mode should be made after DPSIFR1 is cleared to 0x00.

To clear DPSIFR1 to 0x00 after modifying DPSIER1, wait for at least six PCLKB cycles, read DPSIFR1, and then write 0 to DPSIFR1. Six or more PCLKB cycles can be secured, for example, by reading DPSIER1.

DPSIFR1 is not initialized by the internal reset signal (Deep Software Standby reset signal) that is the source to cancel the Deep Software Standby mode. For details, see [section 6, Resets](#).

DIRQnF flag (IRQn-DS Pin Deep Software Standby Cancel Flag) (n = 8 to 15)

The DIRQnF flag indicates that a cancel request by the IRQn-DS pin has been generated.

[Setting condition]

A cancel request by the IRQn-DS pin specified by DPSIEGR1 is generated.

[Clearing condition]

Writing 0 to each flag after 1 is read.

11.2.29 DPSIFR2 : Deep Software Standby Interrupt Flag Register 2

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xA20

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	DNMIF	DRTC AIF	DRTC IIF	DPVD 2IF	DPVD 1IF

Value after reset: 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	DPVD1IF	PVD1 Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ¹
1	DPVD2IF	PVD2 Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ¹

Bit	Symbol	Function	R/W
2	DRTCIIF	RTC Interval Interrupt Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}
3	DRTCAIF	RTC Alarm Interrupt Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}
4	DNMIF	NMI Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}
7:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-4, P-TYPE-2

Note 1. Only 0 can be written to clear flag. The flag must be cleared by writing 0 after 1 is read.

Each flag is set to 1 when a cancel request specified by DPSIEGR2 is generated.

Each flag may be set to 1 when a cancel request is generated in any mode (not even Deep Software Standby mode) or when the setting of DPSIER2 is modified. Therefore, a transition to Deep Software Standby mode should be made after DPSIFR2 is cleared to 0x00.

To clear DPSIFR2 to 0x00 after modifying DPSIER2, wait for at least six PCLKB cycles, read DPSIFR2, and then write 0 to DPSIFR2. Six or more PCLKB cycles can be secured, for example, by reading DPSIER2.

DPSIFR2 is not initialized by the internal reset signal (Deep Software Standby reset signal) that is the source to cancel the Deep Software Standby mode. For details, see [section 6, Resets](#).

DPVDMIF flag (PVDm Deep Software Standby Cancel Flag) (m = 1 to 2)

The DPVDMIF flag indicates that a cancel request by the voltage monitor m signal has been generated.

[Setting condition]

A cancel request is generated by the voltage monitor m signal that is selected in DPSIEGR2.

[Clearing condition]

Writing 0 to each flag after 1 is read.

DRTCIIF flag (RTC Interval Interrupt Deep Software Standby Cancel Flag)

This flag indicates that a cancel request by the RTC interval interrupt signal has been generated.

[Setting condition]

- A cancel request by the RTC interval interrupt signal is generated.

[Clearing condition]

- Writing 0 to each bit after reading the bit as 1.

DRTCAIF flag (RTC Alarm Interrupt Deep Software Standby Cancel Flag)

This flag indicates that a cancel request by the RTC alarm interrupt signal has been generated.

[Setting condition]

- A cancel request by the RTC alarm interrupt signal is generated.

[Clearing condition]

- Writing 0 to each bit after reading the bit as 1.

DNMIF flag (NMI Pin Deep Software Standby Cancel Flag)

This flag indicates that a cancel request by the NMI pin has been generated.

[Setting condition]

- A cancel request by the NMI pin specified by DPSIEGR2 is generated.

[Clearing condition]

- Writing 0 to each bit after reading the bit as 1.

11.2.30 DPSIFR3 : Deep Software Standby Interrupt Flag Register 3

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xA24

Bit position:	7	6	5	4	3	2	1	0
Bit field:	DVBATTADIF	DSOSTDIF	DIWD TIF	—	DULPT1IF	DULPT0IF	—	DUSBFSIF
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DUSBFSIF	USBFS Suspend/Resume Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}
1	—	This bit is read as 0. The write value should be 0.	R/W
2	DULPT0IF	ULPT0 Overflow Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}
3	DULPT1IF	ULPT1 Overflow Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}
4	—	This bit is read as 0. The write value should be 0.	R/W
5	DIWD TIF	IWDT Overflow Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}
6	DSOSTDIF	Sub-clock Oscillation stop detection Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}
7	DVBATTADIF	VBATT Tamper Detection Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}

Note: S-TYPE-4, P-TYPE-2

Note 1. Only 0 can be written to clear flag. The flag must be cleared by writing 0 after 1 is read.

Each flag is set to 1 when the corresponding cancel request is generated.

Each flag may be set to 1 when a cancel request is generated in any mode (not even Deep Software Standby mode) or when the setting of DPSIER3 is modified. Therefore, a transition to Deep Software Standby mode should be made after DPSIFR3 is cleared to 0x00.

To clear DPSIFR3 to 0x00 after modifying DPSIER3, wait for at least six PCLKB cycles, read DPSIFR3, and then write 0 to DPSIFR3. Six or more PCLKB cycles can be secured, for example, by reading DPSIER3.

DPSIFR3 is not initialized by the internal reset signal (Deep Software Standby reset signal) that is the source to cancel the Deep Software Standby mode. For details, see [section 6, Resets](#).

DUSBFSIF flag (USBFS Suspend/Resume Deep Software Standby Cancel Flag)

This flag indicates that a cancel request by the USBFS suspend/resume has been generated.

The DUSBFSIF flag is a flag for USBFS.

[Setting condition]

- A cancel request by the USBFS suspend/resume is generated.

[Clearing condition]

- Writing 0 to each bit after reading the bit as 1.

DULPT0IF flag (ULPT0 Overflow Deep Software Standby Cancel Flag)

This flag indicates that a cancel request by the ULPT0 overflow has been generated.

[Setting condition]

- A cancel request by the ULPT0 is generated.

[Clearing condition]

- Writing 0 to each bit after reading the bit as 1.

DULPT1IF flag (ULPT1 Overflow Deep Software Standby Cancel Flag)

This flag indicates that a cancel request by the ULPT1 overflow has been generated.

[Setting condition]

- A cancel request by the ULPT1 is generated.

[Clearing condition]

- Writing 0 to each bit after reading the bit as 1.

DIWDTIF flag (IWDT Overflow Deep Software Standby Cancel Flag)

This flag indicates that a cancel request by the IWDT overflow has been generated.

[Setting condition]

- A cancel request by the IWDT is generated.

[Clearing condition]

- Writing 0 to each bit after reading the bit as 1.

DSOSTDIF flag (Sub-clock Oscillation stop detection Deep Software Standby Cancel Flag)

This flag indicates that a cancel request by the Sub-clock Oscillation stop detection has been generated.

[Setting condition]

- A cancel request by the SOSTD is generated.

[Clearing condition]

- Writing 0 to each bit after reading the bit as 1.

DVBATTADIF flag (VBATT Tamper Detection Deep Software Standby Cancel Flag)

This flag indicates that a cancel request by the VBATT tamper detection has been generated.

[Setting condition]

- A cancel request by the VBATT tamper detection is generated.

[Clearing condition]

- Writing 0 to each bit after reading the bit as 1.

11.2.31 DPSIFR4 : Deep Software Standby Interrupt Flag Register 4

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xB48

Bit position:	7	6	5	4	3	2	1	0
Bit field:	DIRQ2 3F	DIRQ2 2F	DIRQ2 1F	DIRQ2 0F	DIRQ1 9F	DIRQ1 8F	DIRQ1 7F	DIRQ1 6F

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	DIRQ16F	IRQ16-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}
1	DIRQ17F	IRQ17-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}
2	DIRQ18F	IRQ18-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}
3	DIRQ19F	IRQ19-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}
4	DIRQ20F	IRQ20-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}
5	DIRQ21F	IRQ21-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W
6	DIRQ22F	IRQ22-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}
7	DIRQ23F	IRQ23-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}

Note: S-TYPE-4, P-TYPE-2

Note 1. Only 0 can be written to clear flag. The flag must be cleared by writing 0 after 1 is read.

Each flag is set to 1 when a cancel request specified by DPSIEGR3 is generated.

Each flag may be set to 1 when a cancel request is generated in any mode (not even Deep Software Standby mode) or when the setting of DPSIER4 is modified. Therefore, a transition to Deep Software Standby mode should be made after DPSIFR4 is cleared to 0x00.

To clear DPSIFR4 to 0x00 after modifying DPSIER4, wait for at least six PCLKB cycles, read DPSIFR4, and then write 0 to DPSIFR4. Six or more PCLKB cycles can be secured, for example, by reading DPSIER4.

DPSIFR4 is not initialized by the internal reset signal (Deep Software Standby reset signal) that is the source to cancel the Deep Software Standby mode. For details, see [section 6, Resets](#).

DIRQnF flag (IRQn-DS Pin Deep Software Standby Cancel Flag) (n = 16 to 23)

The DIRQnF flag indicates that a cancel request by the IRQn-DS pin has been generated.

[Setting condition]

A cancel request by the IRQn-DS pin specified by DPSIEGR3 is generated.

[Clearing condition]

Writing 0 to each flag after 1 is read.

11.2.32 DPSIFR5 : Deep Software Standby Interrupt Flag Register 5

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xB4C

Bit position: 7 6 5 4 3 2 1 0

Bit field:

DIRQ3 1F	DIRQ3 0F	DIRQ2 9F	DIRQ2 8F	DIRQ2 7F	DIRQ2 6F	DIRQ2 5F	DIRQ2 4F
-------------	-------------	-------------	-------------	-------------	-------------	-------------	-------------

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	DIRQ24F	IRQ24-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}
1	DIRQ25F	IRQ25-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}
2	DIRQ26F	IRQ26-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}
3	DIRQ27F	IRQ27-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}
4	DIRQ28F	IRQ28-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}
5	DIRQ29F	IRQ29-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}
6	DIRQ30F	IRQ30-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}
7	DIRQ31F	IRQ31-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W ^{*1}

Note: S-TYPE-4, P-TYPE-2

Note 1. Only 0 can be written to clear flag. The flag must be cleared by writing 0 after 1 is read.

Each flag is set to 1 when a cancel request specified by DPSIEGR4 is generated.

Each flag may be set to 1 when a cancel request is generated in any mode (not even Deep Software Standby mode) or when the setting of DPSIER5 is modified. Therefore, a transition to Deep Software Standby mode should be made after DPSIFR5 is cleared to 0x00.

To clear DPSIFR5 to 0x00 after modifying DPSIER5, wait for at least six PCLKB cycles, read DPSIFR5, and then write 0 to DPSIFR5. Six or more PCLKB cycles can be secured, for example, by reading DPSIER5.

DPSIFR5 is not initialized by the internal reset signal (Deep Software Standby reset signal) that is the source to cancel the Deep Software Standby mode. For details, see [section 6, Resets](#).

DIRQnF flag (IRQn-DS Pin Deep Software Standby Cancel Flag) (n = 24 to 31)

The DIRQnF flag indicates that a cancel request by the IRQn-DS pin has been generated.

[Setting condition]

A cancel request by the IRQn-DS pin specified by DPSIEGR4 is generated.

[Clearing condition]

Writing 0 to each flag after 1 is read.

11.2.33 DPSIEGR0 : Deep Software Standby Interrupt Edge Register 0

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xA28

Bit position: 7 6 5 4 3 2 1 0

Bit field:

DIRQ7 EG	DIRQ6 EG	DIRQ5 EG	DIRQ4 EG	DIRQ3 EG	DIRQ2 EG	DIRQ1 EG	DIRQ0 EG
-------------	-------------	-------------	-------------	-------------	-------------	-------------	-------------

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	DIRQ0EG	IRQ0-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
1	DIRQ1EG	IRQ1-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
2	DIRQ2EG	IRQ2-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
3	DIRQ3EG	IRQ3-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
4	DIRQ4EG	IRQ4-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
5	DIRQ5EG	IRQ5-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
6	DIRQ6EG	IRQ6-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
7	DIRQ7EG	IRQ7-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W

Note: S-TYPE-4, P-TYPE-2

DPSIEGR0 is not initialized by the internal reset signal (Deep Software Standby reset signal) that is the source to cancel the Deep Software Standby mode. For details, see [section 6, Resets](#).

11.2.34 DPSIEGR1 : Deep Software Standby Interrupt Edge Register 1

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xA2C

Bit position:	7	6	5	4	3	2	1	0
Bit field:	DIRQ1 5EG	DIRQ1 4EG	DIRQ1 3EG	DIRQ1 2EG	DIRQ1 1EG	DIRQ1 0EG	DIRQ9 EG	DIRQ8 EG

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	DIRQ8EG	IRQ8-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
1	DIRQ9EG	IRQ9-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
2	DIRQ10EG	IRQ10-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
3	DIRQ11EG	IRQ11-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
4	DIRQ12EG	IRQ12-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W

Bit	Symbol	Function	R/W
5	DIRQ13EG	IRQ13-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
6	DIRQ14EG	IRQ14-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
7	DIRQ15EG	IRQ15-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W

Note: S-TYPE-4, P-TYPE-2

DPSIEGR1 is not initialized by the internal reset signal (Deep Software Standby reset signal) that is the source to cancel the Deep Software Standby mode. For details, see [section 6, Resets](#).

11.2.35 DPSIEGR2 : Deep Software Standby Interrupt Edge Register 2

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xA30

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	DNMI EG	—	—	DPVD 2EG	DPVD 1EG

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	DPVD1EG	PVD1 Edge Select 0: A cancel request is generated when VCC < Vdet1 (fall) is detected 1: A cancel request is generated when VCC ≥ Vdet1 (rise) is detected	R/W
1	DPVD2EG	PVD2 Edge Select 0: A cancel request is generated when VCC < Vdet2 (fall) is detected 1: A cancel request is generated when VCC ≥ Vdet2 (rise) is detected	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
4	DNMIEG	NMI Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-4, P-TYPE-2

DPSIEGR2 is not initialized by the internal reset signal (Deep Software Standby reset signal) that is the source to cancel the Deep Software Standby mode. For details, see [section 6, Resets](#).

11.2.36 DPSIEGR3 : Deep Software Standby Interrupt Edge Register 3

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xA34

Bit position:	7	6	5	4	3	2	1	0
Bit field:	DIRQ2 3EG	DIRQ2 2EG	DIRQ2 1EG	DIRQ2 0EG	DIRQ1 9EG	DIRQ1 8EG	DIRQ1 7EG	DIRQ1 6EG

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	DIRQ16EG	IRQ16-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
1	DIRQ17EG	IRQ17-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
2	DIRQ18EG	IRQ18-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
3	DIRQ19EG	IRQ19-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
4	DIRQ20EG	IRQ20-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
5	DIRQ21EG	IRQ21-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
6	DIRQ22EG	IRQ22-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
7	DIRQ23EG	IRQ23-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W

Note: S-TYPE-4, P-TYPE-2

DPSIEGR3 is not initialized by the internal reset signal (Deep Software Standby reset signal) that is the source to cancel the Deep Software Standby mode. For details, see [section 6, Resets](#).

11.2.37 DPSIEGR4 : Deep Software Standby Interrupt Edge Register 4

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xB50

Bit position: 7 6 5 4 3 2 1 0

Bit field:	DIRQ3 1EG	DIRQ3 0EG	DIRQ2 9EG	DIRQ2 8EG	DIRQ2 7EG	DIRQ2 6EG	DIRQ2 5EG	DIRQ2 4EG
------------	--------------	--------------	--------------	--------------	--------------	--------------	--------------	--------------

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	DIRQ24EG	IRQ24-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
1	DIRQ25EG	IRQ25-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
2	DIRQ26EG	IRQ26-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
3	DIRQ27EG	IRQ27-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
4	DIRQ28EG	IRQ28-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W

Bit	Symbol	Function	R/W
5	DIRQ29EG	IRQ29-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
6	DIRQ30EG	IRQ30-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
7	DIRQ31EG	IRQ31-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W

Note: S-TYPE-4, P-TYPE-2

DPSIEGR4 is not initialized by the internal reset signal (Deep Software Standby reset signal) that is the source to cancel the Deep Software Standby mode. For details, see [section 6, Resets](#).

11.2.38 SYOCDCCR : System Control OCD Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xA38

Bit position: 7 6 5 4 3 2 1 0

Bit field:	DBGEN	—	—	—	—	—	—
------------	-------	---	---	---	---	---	---

Value after reset: 1 0 0 0 0 0 0 x

Bit	Symbol	Function	R/W
0	—	The read value is undefined. The write value should be 0.	R/W
6:1	—	These bits are read as 0. The write value should be 0.	R/W
7	DBGEN	Debugger Enable bit 0: Disable on-chip debugger function 1: Enable on-chip debugger function*1	R/W

Note: S-TYPE-5 P-TYPE-2

Note 1. The value of DBGEN can be set by software. However, it is not until the debugger authentication passes that the on-chip debugger function is valid. That is, even if the DBGEN is 1, the on-chip debugger function is invalid unless the debugger authentication passes. For detail, see [section 2, CPU](#).

This register is not controlled by any security attribute register (e.g. LPMSAR, DPFSAR).

SYOCDCCR is not initialized by the internal reset signal (Deep Software Standby reset signal) that is the source to cancel the Deep Software Standby mode.

DBGEN bit (Debugger Enable bit)

The DBGEN enables the on-chip debugger function.

[Setting condition]

- Power-on reset is generated
- Voltage Monitor 0 reset is generated.
- Core Voltage Monitor reset is generated.
- Writing 1 to the bit.

[Clearing condition]

- Writing 0 to the bit

11.2.39 PLL1LDOCR : PLL1-LDO Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xB04

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	SKEE P	LDOS TP
Value after reset:	0	0	0	0	0	x	0	0

Bit	Symbol	Function	R/W
0	LDOSTP	LDO Stop 0: PLL1-LDO is enabled 1: PLL1-LDO is stopped	R/W
1	SKEEP	STBY Keep 0: PLL1-LDO is stopped during Software Standby mode. 1: PLL1-LDO state before Software Standby mode is retained during Software Standby mode.	R/W
2	—	The read value is undefined.	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

You can control the operation of the LDO dedicated for PLL1 (PLL1-LDO) by using this register.

LDOSTP bit (LDO Stop)

The PLL1-LDO state during Normal mode is controlled by this bit.

After setting 1 to this bit, do not set 0 to this bit for at least 1 μ s. This bit can be set 0 to this bit only when 1 μ s has elapsed after setting 1 to this bit.

When this bit is changed from 1 to 0, it takes 25 μ s for the power supply to stabilize.

Allow the power supply to stabilize before operating the PLL1.

SKEEP bit (STBY Keep)

The PLL1-LDO state during Software Standby mode is controlled by this bit. If the PLL1-LDO operation is stopped during Software Standby mode, the power consumption during Software Standby mode can be reduced.

11.2.40 PLL2LDOCR : PLL2-LDO Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xB08

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	SKEE P	LDOS TP
Value after reset:	0	0	0	0	0	x	0	0

Bit	Symbol	Function	R/W
0	LDOSTP	LDO Stop 0: PLL2-LDO is enabled 1: PLL2-LDO is stopped	R/W
1	SKEEP	STBY Keep 0: PLL2-LDO is stopped during Software Standby mode. 1: PLL2-LDO state before Software Standby mode is retained during Software Standby mode.	R/W
2	—	The read value is undefined.	R/W

Bit	Symbol	Function	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

You can control the operation of the LDO dedicated for PLL2 (PLL2-LDO) by using this register.

LDOSTP bit (LDO Stop)

The PLL2-LDO state during Normal mode is controlled by this bit.

After setting 1 to this bit, do not set 0 to this bit for at least 1 μ s. This bit can be set 0 to this bit only when 1 μ s has elapsed after setting 1 to this bit.

When this bit is changed from 1 to 0, it takes 25 μ s for the power supply to stabilize.

Allow the power supply to stabilize before operating the PLL2.

SKEEP bit (STBY Keep)

The PLL2-LDO state during Software Standby mode is controlled by this bit. If the PLL2-LDO operation is stopped during Software Standby mode, the power consumption during Software Standby mode can be reduced.

11.2.41 HOCOLDOCR : HOCO-LDO Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xB0C

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	SKEE P	LDOS TP

Value after reset: 0 0 0 0 0 0 x 0 0

Bit	Symbol	Function	R/W
0	LDOSTP	LDO Stop 0: HOCO-LDO is enabled 1: HOCO-LDO is stopped	R/W
1	SKEEP	STBY Keep 0: HOCO-LDO is stopped during Software Standby mode. 1: HOCO-LDO state before Software Standby mode is retained during Software Standby mode.	R/W
2	—	The read value is undefined.	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

You can control the operation of the LDO dedicated for HOCO (HOCO-LDO) by using this register.

LDOSTP bit (LDO Stop)

The HOCO-LDO state during Normal mode is controlled by this bit.

After setting 1 to this bit, do not set 0 to this bit for at least 2 μ s. This bit can be set 0 to this bit only when 2 μ s has elapsed after setting 1 to this bit.

When this bit is changed from 1 to 0, it takes 5 μ s for the power supply to stabilize.

Allow the power supply to stabilize before operating the HOCO.

SKEEP bit (STBY Keep)

The HOCO-LDO state during Software Standby mode is controlled by this bit. If the HOCO-LDO operation is stopped during Software Standby mode, the power consumption during Software Standby mode can be reduced.

11.2.42 VSCR : Voltage Scaling Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x014

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	VSCM TSF	—	VSCM[2:0]		
Value after reset:	0	0	0	0	0	0	1	0

Bit	Symbol	Function	R/W
2:0	VSCM[2:0]	Voltage Scaling Control Mode Bit Set the target voltage of VDD. 0 0 1: VSCR_1 0 1 0: VSCR_2 (default) Setting other than above are prohibited.	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
4	VSCMTSF	Voltage Scaling Control Mode Transition Status Flag - Read 0: Transition completed 1: During transition - Write - The write value should be 0.	R
7:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

The VSCR register controls the voltage scaling operation.

VSCM[2:0] bits (Voltage Scaling Control Mode Bit)

The VSCM bits specify the target voltage of VDD for voltage scaling operation.

VSCMTSF flag (Voltage Scaling Control Mode Transition Status Flag)

The VSCMTSF flag indicates the voltage scaling transition state when the operating voltage scaling control is executed. When a write access is attempted to change the voltage scaling control mode, the VSCMTSF flag is set to 1. The flag becomes 0 after a transition to the voltage scaling control mode is completed. Make sure that the VSCMTSF flag is 0 (completed operating voltage scaling control mode transition) before the next processing.

11.2.43 SVSCR : SSTBY Voltage Scaling Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xA9C

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	SVSCM[2:0]		
Value after reset:	0	0	0	0	0	0	1	0

Bit	Symbol	Function	R/W
2:0	SVSCM[2:0]	SSTBY Voltage Scaling Control Mode Bit Set the target voltage of VDD in Software Standby mode. 0 0 1: SVSCR_1 0 1 0: SVSCR_2 (default) 0 1 1: SVSCR_3 1 0 0: SVSCR_4 1 0 1: SVSCR_5 Setting other than above are prohibited.	R/W

Bit	Symbol	Function	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

The SVSCR register controls the voltage scaling operation in Software Standby mode.

SVSCM[2:0] bits (SSTBY Voltage Scaling Control Mode Bit)

The SVSCM bits specify the target voltage of VDD for voltage scaling operation in Software Standby mode.

It is forbidden to set SVSCR_1 and SS2LP[1:0] = 01b and transition to Software Standby mode.

When SVSCR_3, SVSCR_4 or SVSCR_5 is set, the interrupt factor that cancels Software Standby mode is limited to NMI pin or IRQ pin.

11.2.44 CPUDSCR : CPU Deep Sleep Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x100

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	PGD1	PGD0
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PGD0	Power Gating Disable for CPU0 0: Power Gating Enable during CPU0 Deep Sleep (default) 1: Power Gating Disable during CPU0 Deep Sleep	R/W
1	PGD1	Power Gating Disable for CPU1 0: Power Gating Enable during CPU1 Deep Sleep (default) 1: Power Gating Disable during CPU1 Deep Sleep	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

The CPUDSCR register controls the power gating during CPU deep sleep mode.

Each bit of CPUDSCR should be changed while the target CPU is operation.

When transitioning to Software Standby mode and Deep Software Standby mode, CPUDSCR.PGDn must be set to 0.

PGD0 bit (Power Gating Disable for CPU0)

The PGD0 bit specifies the power gating control of CPU0 power domain during CPU0 deep sleep mode.

PGD1 bit (Power Gating Disable for CPU1)

The PGD1 bit specifies the power gating control of CPU1 power domain during CPU1 deep sleep mode.

11.3 Reducing Power Consumption by Switching Clock Signals

The clock frequency changes when the SCKDIVCR and SCKDIVCR2 registers are set.

For information on module and clock associations, see [section 9.2.2. SCKDIVCR : System Clock Division Control Register](#) and [section 9.2.3. SCKDIVCR2 : System Clock Division Control Register 2](#).

11.4 Module-Stop Function

The module-stop function can be set for each on-chip peripheral module.

When the MSTPmi bit (m = A to E, i = 31 to 0) in MSTPCRA to MSTPCRE is set to 1, the corresponding module stops operating and enters the module-stop state, but the CPU continues to operate independently. Clearing the MSTPmi bit to 0 cancels the module-stop state, allowing the module to restart operating at the end of the bus cycle. The internal states of modules are retained in the module-stop state.

After a reset is canceled, all modules other than the DMAC, DTC and SRAMn ($n = 0$ to 3) are placed in the module-stop state. Do not access the module while the corresponding MSTPmi bit is 1. Otherwise, the read/write data nor the operation of the module is not guaranteed. Also, do not set 1 to the MSTPmi bit while the corresponding module is being accessed.

If the CPUCLK0 is faster than the ICLK maximum frequency in the electrical characteristics, after the MSTPmi bit is changed, a wait time should be provided according to the flow shown in [Figure 11.2](#). It is recommended to use software to measure the wait time. Be sure to consider the worst-case conditions to ensure that the required wait time elapses. If an interrupt is unavoidably generated during the wait time, retry the measurement after return from the interrupt.

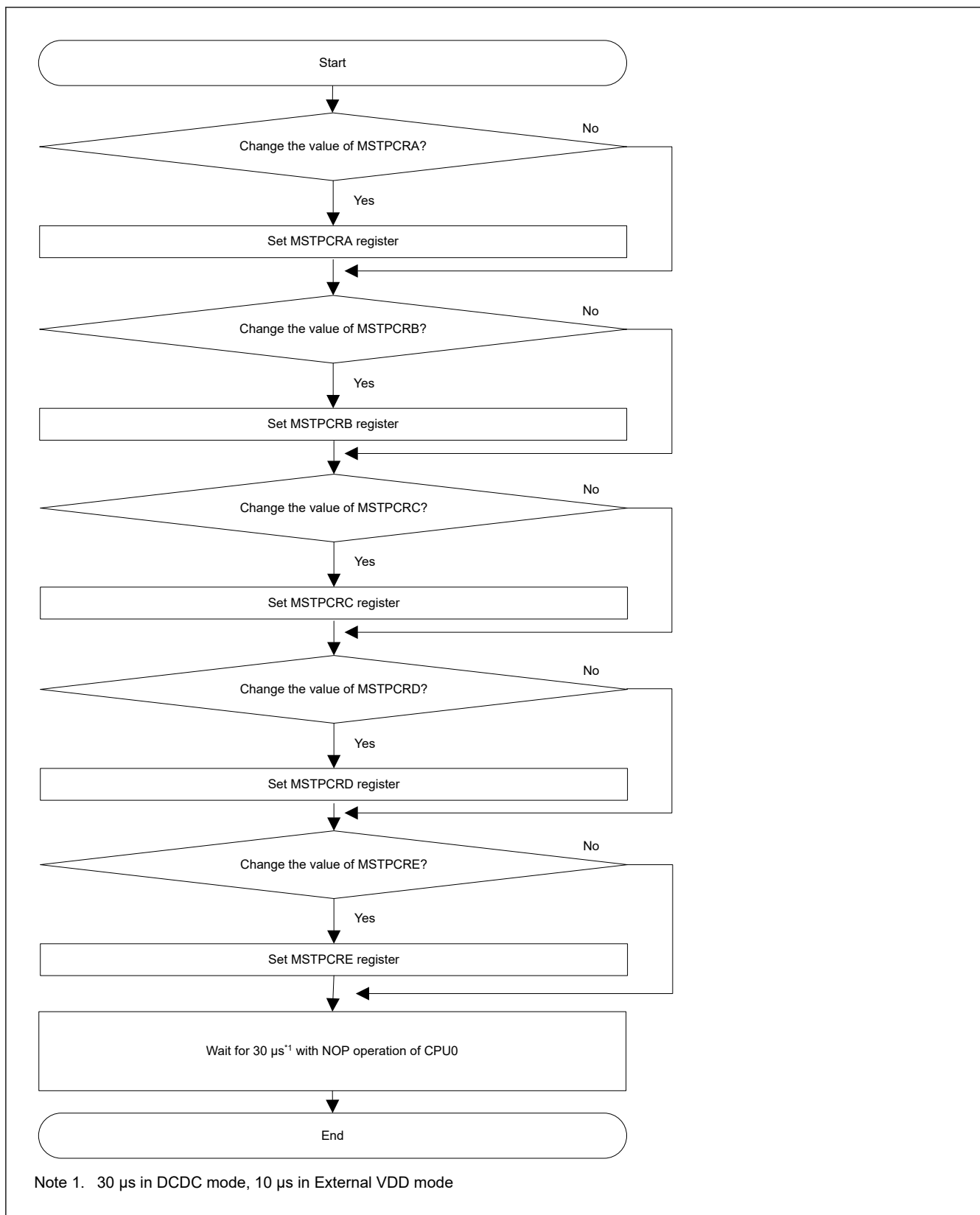


Figure 11.2 Setting flow of module stop control registers in case of CPUCLK0 is higher than the ICLK maximum frequency in electrical characteristics

11.5 Power Gating Control

The MCU supports the power gating function of the power domain.

User can individually control the power on/off of the divided power domains.

In the power gating state, the domain is powered off and the state is the same as SSTBY in [Table 11.3](#).

11.5.1 Power Gating Control Using Power Domain Control Register

[Table 11.7](#) shows the relationship PDCTR register and the target power domain.

When the PDCTR register is set, power gating control is executed. In addition, when using the power gating function, it should be set MOCOER.MCSTP to 0 (MOCO is operated) in advance.

Make sure that the power domain control of the CPU and the power domain control by the PDCTRESWM do not conflict.

Table 11.7 Relationship between the PDCTR register and the target power domain

Control register				Target Power Domain name	Implemented IP name
Register name	PDDE	PDCSF	PDPGSF		
PDCTRESWM	Available	Available	Available	ESWM domain	ESWM

Examples of the procedures for power gating control:

1. Power OFF by PDCTRn.PDDE*¹

STEP1 : Confirm that the target domain is not performing the Power gating process

- Check PDCTRn.PDCSF = 0

STEP2 : Set the module stop control register to stop.

- Write MSTPCR* = 1 (*: all register for target domain)

STEP3 : Set PDCTRn.PDDE bit and start the power control.

- Write PDCTRn.PDDE = 1

STEP4 : Confirm that the power gating process is completed and the powered off

- Check PDCTRn.PDCSF = 0 & PDCTRn.PDPGSF = 1

2. Power ON by PDCTRn.PDDE*¹

STEP1 : Confirm that the target domain is not performing the Power gating process

- Check PDCTRn.PDCSF = 0

STEP2 : Set PDCTRn.PDDE bit and start the power control.

- Write PDCTRn.PDDE = 0

STEP3 : Confirm that the power gating process is completed and the powered on

- Check PDCTRn.PDCSF = 0 & PDCTRn.PDPGSF = 0

STEP4 : Set the module stop control register to operate.

- Write MSTPCR* = 0 (*: all register for target domain)

Note 1. When PDCTRn.PDCSF bit is "1" (during power gating control processing), PDCTRn register must not be changed.

11.6 Low Power Modes

11.6.1 CPU Sleep Mode and CPU Deep Sleep Mode

See [section 2.7. Sleep Modes for Low Power States](#)

11.6.2 Software Standby Mode

11.6.2.1 Entering Software Standby Mode

When WFI instructions of both CPUs are executed while LPSCR.LPMD bit is 0x5 and CPU_n.SCR.SLEEPDEEP bit is 1, the MCU enters Software Standby mode depending on LPSCR.LPMD setting. When transitioning to Software Standby mode, CPUDSCR.PGD_n must be set to 0. It must be set MOCO_{CR}.MCSTP to 0 (MOCO is operated) before executed the WFI instruction to entry this mode. Take care that the transition is suspended during the configuration update transfer.

In Software Standby mode, the CPU, most of the on-chip peripheral functions and most of the oscillators stop^{*1}. However, the contents of the CPU internal registers and RAM data, the states of on-chip peripheral functions and the I/O ports are retained. Software Standby mode allows significant reduction in power consumption because most of the oscillators stops^{*1}. [Table 11.3](#) shows the status of each on-chip peripheral functions and oscillators. In addition, Software Standby mode allows the internal regulator to transition into a low power state through SSCR1.SS2LP register. This further reduces power consumption. Available resets or interrupts in Software Standby mode make the MCU to return to Normal mode. See [Table 11.4](#) for available interrupt sources for each mode. In case of using an interrupt to return to Normal mode, corresponding IELSR_n register must be set before executing a WFI instruction. For the detail, see [section 14, Interrupt Controller Unit \(ICU\)](#).

Note 1. The state of the oscillators depends on the setting of the control register for each oscillator. For details, see [section 9, Clock Generation Circuit](#).

The status of address bus and bus control signals in Software Standby mode can be selected by SBYCR.OPE bit. The SBYCR.OPE bit specifies whether to set to the high-impedance state or to retain the output of the address bus and bus control signals in Software Standby mode.

Clear DMAST.DMST bit and DTCST.DTCST bit to 0 before executing WFI instruction.

Counting by the IWD_T stops when the MCU enters Software Standby mode while the IWD_T is in auto-start mode and the OFS0.IWD_TSTPCTL bit is 1 (IWD_T stops in Sleep or Software Standby).

Counting by the IWD_T continues when the MCU enters Software Standby mode while the IWD_T is in auto-start mode and the OFS0.IWD_TSTPCTL bit is 0 (IWD_T does not stop in Sleep or Software Standby).

Counting by the IWD_T stops when the MCU enters Software Standby mode while the IWD_T is in register start mode and the SLCSTP bit in IWD_TCSTPR is 1 (IWD_T stops in Sleep or Software Standby).

Counting by the IWD_T continues when the MCU enters Software Standby mode while the IWD_T is in register start mode and the SLCSTP bit in IWD_TCSTPR is 0 (IWD_T does not stop in Sleep or Software Standby).

WDT stops counting when the MCU enters Software Standby mode because the PCLK stops.

Do not enter Software Standby mode while the MRAM is in programming procedure. In order to enter Software Standby mode, execute a WFI instruction after programming procedure completes.

In External VDD mode, do not enter Software Standby Mode.

[Table 11.8](#) shows the setting of the related control bits and the modes to enter after executing WFI instruction.

Table 11.8 Setting of the bits what affect the mode after executing WFI instruction (1 of 2)

		Setting mode of LPSCR.LPMD bit and CPU _n .SCR.SLEEPDEEP (n = 0, 1) bit			
		Software Standby (LPSCR.LPMD = 0x5, CPU _n .SCR. SLEEPDEEP = 1)	Deep Software Standby 1 (LPSCR.LPMD = 0x8, CPU _n .SCR. SLEEPDEEP = 1)	Deep Software Standby 2 (LPSCR.LPMD = 0x9, CPU _n .SCR. SLEEPDEEP = 1)	Deep Software Standby 3 (LPSCR.LPMD = 0xA, CPU _n .SCR. SLEEPDEEP = 1)
MENTRYR.MENTRY	0	Software Standby	Deep Software Standby 1	Deep Software Standby 2	Deep Software Standby 3
	1	CPU Deep sleep mode	CPU Deep sleep mode	CPU Deep sleep mode	CPU Deep sleep mode
CPUDSCR.PGD0	0	Software Standby	Deep Software Standby 1	Deep Software Standby 2	Deep Software Standby 3
	1	CPU Deep sleep mode	CPU Deep sleep mode	CPU Deep sleep mode	CPU Deep sleep mode

Table 11.8 Setting of the bits what affect the mode after executing WFI instruction (2 of 2)

		Setting mode of LPSCR.LPMD bit and CPUUn.SCR.SLEEPDEEP (n = 0, 1) bit			
		Software Standby (LPSCR.LPMD = 0x5, CPUUn.SCR. SLEEPDEEP = 1)	Deep Software Standby 1 (LPSCR.LPMD = 0x8, CPUUn.SCR. SLEEPDEEP = 1)	Deep Software Standby 2 (LPSCR.LPMD = 0x9, CPUUn.SCR. SLEEPDEEP = 1)	Deep Software Standby 3 (LPSCR.LPMD = 0xA, CPUUn.SCR. SLEEPDEEP = 1)
CPUDSCR.PGD1	0	Software Standby	Deep Software Standby 1	Deep Software Standby 2	Deep Software Standby 3
	1	CPU Deep sleep mode	CPU Deep sleep mode	CPU Deep sleep mode	CPU Deep sleep mode
OFS0.IWDTSTPCTL (auto-start mode)	0	Software Standby	Deep Software Standby 1	Deep Software Standby 1	Deep Software Standby 1
	1			Deep Software Standby 2	Deep Software Standby 3
IWDTCSSTPR.SLCSTP (register start mode)	0	Software Standby	Deep Software Standby 1	Deep Software Standby 1	Deep Software Standby 1
	1			Deep Software Standby 2	Deep Software Standby 3
OFS2.CVMRDIS	0	Software Standby	Software Standby	Software Standby	Software Standby
	1		Deep Software Standby 1	Deep Software Standby 2	Deep Software Standby 3
PVD1CR0.RI	0	Software Standby	Deep Software Standby 1	Deep Software Standby 2	Deep Software Standby 3
	1			Deep Software Standby 1	Deep Software Standby 1
PVD2CR0.RI	0	Software Standby	Deep Software Standby 1	Deep Software Standby 2	Deep Software Standby 3
	1			Deep Software Standby 1	Deep Software Standby 1
PVD4CR0.RE	0	Software Standby	Deep Software Standby 1	Deep Software Standby 2	Deep Software Standby 3
	1				Deep Software Standby 2 ^{*1}
PVD5CR0.RE	0	Software Standby	Deep Software Standby 1	Deep Software Standby 2	Deep Software Standby 3
	1				Deep Software Standby 2 ^{*1}

Note 1. When at least PVDnCR0.RE (n = 4, 5) bit is 1 and all PVDnCR0.RI (n = 1, 2) bit are 0 and OFS0.IWDTSTPCTL is 1 and IWDTCSSTPR.SLCSTP is 1, the transition is to Deep Software Standby mode 2 even if LPSCR is set to 0xA.

11.6.2.2 Canceling Software Standby Mode

Software Standby mode is canceled by an available interrupt shown in Table 11.4, RES pin reset, a power-on reset, a voltage monitor reset, a core voltage monitor reset, a reset caused by IWDt underflow or a temperature monitor reset.

The oscillators which were operating before the transition to Software Standby mode restarts. After the oscillation of all these oscillators has been stabilized, the MCU returns from Software Standby mode to Normal mode.

1. Canceling by an interrupt

When an available interrupt request (for available interrupt, See Table 11.4) is generated, each oscillator which was operating before the transition to Software Standby mode restarts. After all these oscillators have been stabilized, the MCU returns from Software Standby mode to Normal mode and starts interrupt handling. When canceling the software standby mode by interrupt, each power domain returns to the same power state as the state before entering the software standby state.

2. Canceling by a RES pin reset

When RES pin is driven low, the MCU enters the reset state and oscillators, whose default status are operating, starts the oscillation. Make sure to keep RES pin low for the time period specified in [section 60, Electrical Characteristics](#). When RES pin is driven high after the specified time period, the CPU starts the reset exception handling.

3. Canceling by a power-on reset

Software Standby mode is canceled by a power-on reset and the MCU starts the reset exception handling.

4. Canceling by a voltage monitor reset

Software Standby mode is canceled by a voltage monitor reset from the voltage detection circuit and the MCU starts the reset exception handling.

5. Canceling by IWDG reset

Software Standby mode is canceled by an internal reset generated by IWDG underflow and the MCU starts the reset exception handling. However, IWDG stops in Software Standby mode and an internal reset for Canceling Software Standby mode is not generated in the following conditions:

- $OFS0.IWDGSTRT = 0$ and $OFS0.IWDGSTPCTL = 1$ (auto-start mode)
- $IWDGCTPR.SLCSTP = 1$ (register start mode)

6. Canceling by a core voltage monitor reset

Software Standby mode is canceled by a core voltage monitor reset (CVMR) and the MCU starts the reset exception handling.

7. Canceling by a temperature monitor reset

Software Standby mode is canceled by a temperature monitor reset and the MCU starts the reset exception handling.

11.6.2.3 Example of Software Standby Mode Application

[Figure 11.3](#) shows an example of entry to Software Standby mode by a falling edge of IRQn pin, and exit from Software Standby mode by a rising edge of IRQn pin.

In this example, an IRQn pin interrupt is accepted with $IRQCRi.IRQMD[1:0]$ ($i = 0$ to 31) bits of the ICU are 00b (falling edge) in Normal mode, and then $IRQCRi.IRQMD[1:0]$ ($i = 0$ to 31) bits are set to 01b (rising edge). Then, if MOCO is not operated, it must set $MOCOCR.MCSTP$ to 0 (MOCO is operated). After that, $LPSCR.LPMD$ bit is set to 0x5 and $CPU_n.SCR.SLEEPDEEP$ bit is set to 1, and then WFI instruction is executed. Thus, entry to Software Standby mode is completed. After that, exit from Software Standby mode is initiated by a rising edge of IRQn pin.

Setting the ICU is also required to exit Software Standby mode. For details, see [section 14, Interrupt Controller Unit \(ICU\)](#). The oscillation stabilization time in [Figure 11.2](#) is specified in [section 60, Electrical Characteristics](#).

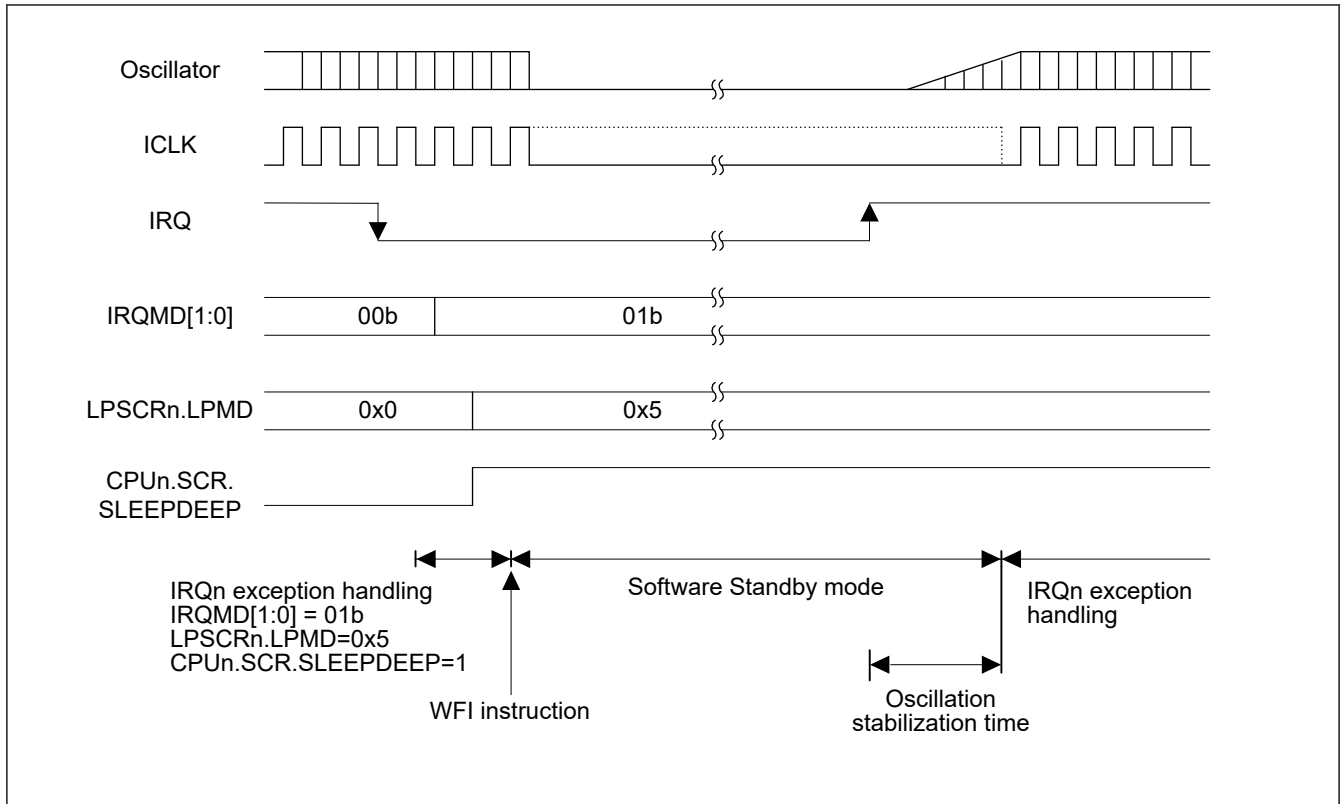


Figure 11.3 Example of Software Standby mode application

11.6.3 Deep Software Standby Mode

11.6.3.1 Transition to Deep Software Standby Mode

When WFI instructions of both CPUs are executed while LPSCR.LPMD bit is 0x8, 0x9, 0xA and CPUUn.SCR.SLEEPDEEP bit is 1, the MCU enters Deep Software Standby mode 1 or 2 or 3 depending on LPSCR.LPMD setting. When transitioning to Deep Software Standby mode, CPUDSCR.PGDn must be set to 0. It should be set MOCOCR.MCSTP to 0 (MOCO is operated) before executed the WFI instruction to entry this mode. See [Table 11.8](#) for the setting of the related control bits.

In these modes, the CPU, on-chip peripheral functions, SRAM, and oscillators are almost stopped as shown in [Table 11.3](#). Furthermore, since the internal power supply to these inactive modules is stopped, power consumption is remarkably reduced. The contents of all the registers of the CPU and specific internal peripheral modules become undefined. [Table 11.3](#) shows the status of each on-chip peripheral functions and oscillators.

In Deep Software Standby mode 1, when OFS1.PVDAS = 0 and OFS1.PVDLPSEL = 0, the battery power supply switch is controlled by the voltage monitor 0 which has lower power consumption than VCC voltage drop detection circuit dedicated for VBATT, so power consumption is further reduced at the case that VBATT function is enabled. In Deep Software Standby mode 2, the internal supply of power to the USB resume detecting unit and IWDT is cut off, reducing power consumption. When OFS1.PVDAS = 0 and OFS1.PVDLPSEL = 0, the battery power supply switch is controlled by the voltage monitor 0 which has lower power consumption than VCC voltage drop detection circuit dedicated for VBATT, so power consumption is further reduced at the case that VBATT function is enabled. In Deep Software Standby mode 3, the PVD is also stopped. For details, see [section 60, Electrical Characteristics](#).

Clear DMAST.DMST bit and DTCST.DTCST bit to 0 before executing WFI instruction.

When the MCU enters Deep Software Standby mode 2 or 3 while the IWDT is in auto-start mode and the OFS0.IWDTSTPCTL bit is 1, power supply to the IWDT-dedicated clock and the IWDT is cut off. Counting by the IWDT also stops.

When the IWDT is in auto-start mode and the OFS0.IWDTSTPCTL bit is 0, the MCU enters Deep Software Standby mode 1 instead of Deep Software Standby Mode 2 or 3, regardless of the setting in the LPSCR.LPMD bit.

When the IWDT is in register start mode and the IWDTCSTPR.SLCSTP is 0, the MCU enters Deep Software Standby mode 1 instead of Deep Software Standby Mode 2 or 3, regardless of the setting in the LPSCR.LPMD.

Do not enter Deep Software Standby mode while the MRAM is in programming procedure. In order to enter Deep Software Standby mode, execute a WFI instruction after programming procedure completes.

In External VDD mode, do not enter Deep Software Standby Mode.

When PVDnCR0.RI ($n = 1, 2$) = 1 (selecting the voltage monitor n reset), the MCU enters Deep Software Standby mode 1 instead of Deep Software Standby mode 2 or 3, regardless of the setting in the LPSCR.LPMD. The I/O port states are same as Software Standby mode.

When at least PVDnCR0.RE ($n = 4, 5$) bit is 1 and all PVDnCR0.RI ($n = 1, 2$) bit are 0 and OFS0.IWDTSTPCTL is 1 and IWDTCSSTPR.SLCSTP is 1, the MCU enters Deep Software Standby mode 2 instead of Deep Software Standby mode 3, regardless of the setting in the LPSCR.LPMD. The I/O port states are same as Software Standby mode.

When the Core Voltage Detection Circuit is enabled (Core Voltage Monitor disable bit (CVMRDIS) in Option Function Select Register 0 (OFS2) is 0), the MCU enters Software Standby mode instead of Deep Software Standby Mode 1, 2 or 3, regardless of the setting in the LPSCR.LPMD bit.

11.6.3.2 Canceling Deep Software Standby Mode

Deep Software Standby mode is canceled by the interrupt shown in [Table 11.4](#), RES pin reset, a power-on reset, a voltage monitor reset, a core voltage monitor reset, a reset caused by IWDTCSTPR underflow or a temperature monitor reset.

(1) Canceling by an interrupt

Canceling by interrupts is controlled by DPSIERn ($n = 0$ to 5) and DPSIFRn ($n = 0$ to 5). When a Deep Software Standby Canceling interrupt is generated, the corresponding flag in DPSIFRn is set to 1. If the interrupt is enabled in DPSIERn, Deep Software Standby mode is canceled. Rising edge or falling edge can be selected by DPSIEGRn ($n = 0$ to 4). The interrupts for which an edge can be selected are the NMI, IRQ0-DS to IRQ31-DS, voltage monitor n ($n = 1, 2$) interrupts. When a Deep Software Standby mode canceling request occurs, the internal power is supplied and MOCO starts oscillating, and then the internal reset (Deep Software Standby reset) is generated for the entire MCU.

The stable MOCO clock is then supplied to the entire MCU and Deep Software Standby reset is canceled, and the MCU starts reset exception handling.

When Deep Software Standby mode is canceled by an external interrupt pin or internal interrupt signal, the RSTSR0.DPSRSTF flag is set to 1.

(2) Canceling by RES pin reset

When the RES pin is driven low, the MCU cancels Deep Software Standby mode and enters the reset state. Make sure to keep RES pin low for the time period specified in [section 60, Electrical Characteristics](#). When RES pin is driven high after the specified time period, the CPU starts the reset exception handling.

(3) Canceling by a power-on reset

Deep Software Standby mode is canceled by a power-on reset and the MCU starts the reset exception handling.

(4) Canceling by a voltage monitor 0 reset

Deep Software Standby mode is canceled by a voltage monitor 0 reset from the voltage detection circuit and the MCU starts the reset exception handling.

(5) Canceling by a voltage monitor n reset ($n = 1, 2, 4, 5$)

Deep Software Standby mode 1 is canceled by a voltage monitor n reset ($n = 1, 2, 4, 5$) from the voltage detection circuit and the MCU starts the reset exception handling.

(6) Canceling by a IWDTCSTPR reset

Deep Software Standby mode 1 is canceled by a IWDTCSTPR reset (an internal reset generated by IWDTCSTPR underflow) and the MCU starts the reset exception handling.

However, IWDTCSTPR stops in Deep Software Standby mode 1 and a IWDTCSTPR reset for Canceling Deep Software Standby mode 1 is not generated in the following conditions:

- OFS0.IWDTSTRT = 0 and OFS0.IWDTSTPCTL = 1 (auto-start mode)
- IWDTCSSTPR.SLCSTP = 1 (register start mode)

(7) Canceling by a temperature monitor reset

Deep Software Standby mode is canceled by a temperature monitor reset and the MCU starts the reset exception handling.

11.6.3.3 Pin States when Deep Software Standby Mode is Canceled

In Deep Software Standby mode, the I/O ports retain the same states as before transition to the Deep Software Standby mode. The MCU is initialized by an internal reset generated when Deep Software Standby mode is canceled. Upon cancellation of Deep Software Standby mode, the reset exception handling starts immediately.

Whether to initialize the I/O ports or to retain the I/O port states as before transition to the Deep Software Standby mode can be selected by the DPSBYCR.IOKEEP bit.

- When the DPSBYCR.IOKEEP bit = 0
I/O ports are initialized by an internal reset generated when Deep Software Standby mode is canceled.
- When the DPSBYCR.IOKEEP bit = 1

Although the MCU is initialized by an internal reset generated when Deep Software Standby mode is canceled, I/O ports retain their states as before transition to the Deep Software Standby mode regardless of the MCU internal state. At this time, the I/O port states remain unchanged as before transition to the Deep Software Standby mode even if settings of I/O ports or peripheral modules are made. Then, the retained I/O port states are released by clearing the DPSBYCR.IOKEEP bit to 0, and the MCU operates according to the internal state.

The DPSBYCR.IOKEEP bit is not initialized by the internal reset signal (Deep Software Standby reset signal) that is the source to cancel the Deep Software Standby mode.

11.6.3.4 Example of Deep Software Standby Mode Application

(1) Entering and exiting Deep Software Standby mode

Figure 11.4 shows an example where a transition to Deep Software Standby mode is made at the falling edge of the IRQn-DS pin, and exiting Deep Software Standby mode is made at the rising edge of the IRQn-DS pin. In this example, an IRQn interrupt is accepted with the IRQCRi.IRQMD[1:0] (i = 0 to 31) bits of the ICU set to 00b (falling edge). Then, after the DPSIEGRy.DIRQnEG (y = 0 or 1 or 4 or 5, n = 0 to 31) bit is set to 1 (rising edge) and the LPSCR.LPMD bit is set to 0x8 or 0x9 or 0xA and CPUUn.SCR.SLEEPDEEP bit is set to 1, the WFI instruction is executed. Thus, a transition to Deep Software Standby mode is made. After that, exiting Deep Software Standby mode is made at the rising edge of the IRQ-DS pin.

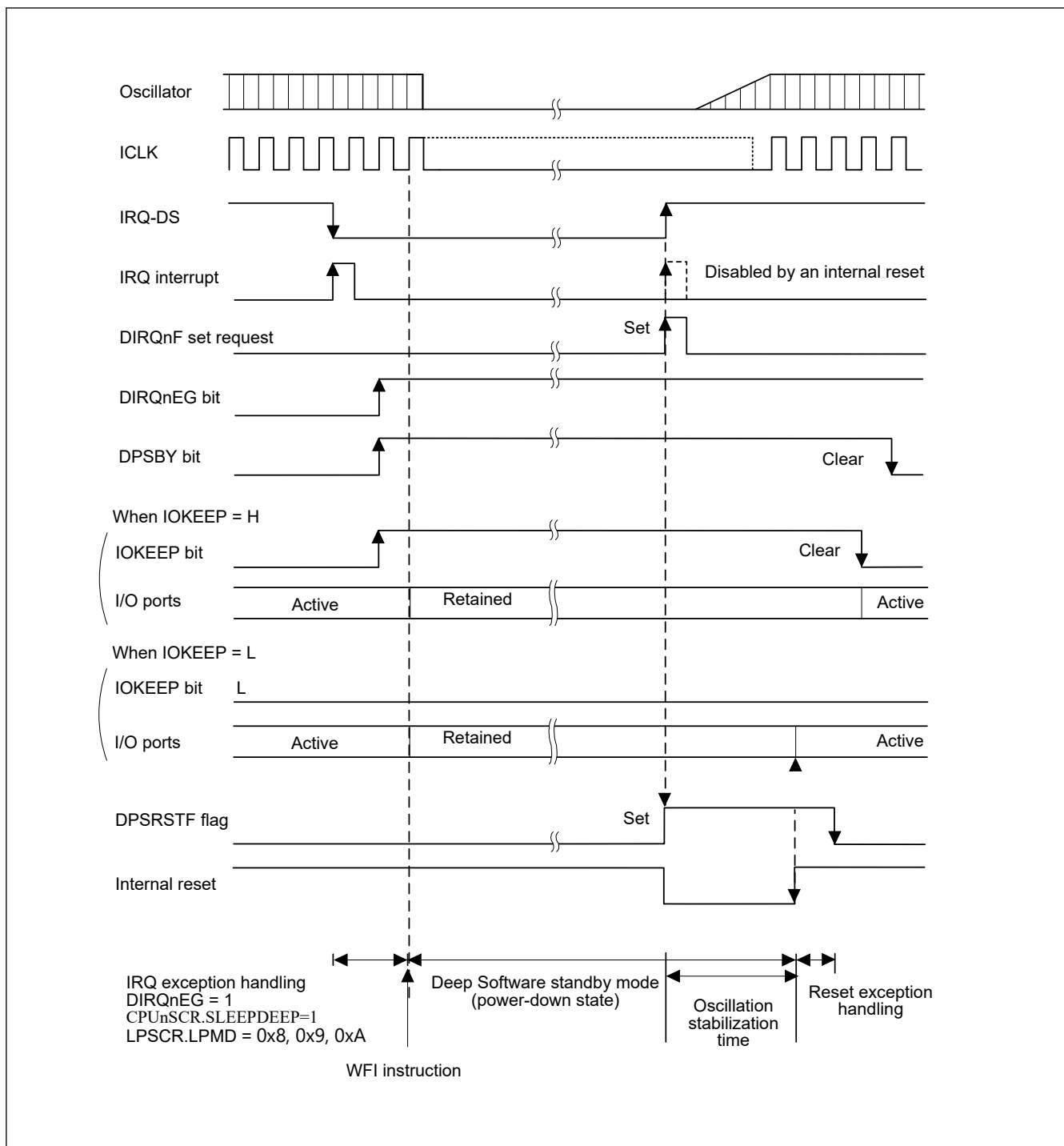


Figure 11.4 Example of Deep Software Standby mode application

11.6.3.5 Usage Flow for Deep Software Standby Mode

Figure 11.5 shows an example of a flowchart to use Deep Software Standby mode.

In this example, the RSTSR0.DPSRSTF flag of the reset function is read after the reset exception handling to determine whether a reset was generated by the RES pin or by the cancellation of Deep Software Standby mode.

In the case of a reset by the RES pin, a transition to Deep Software Standby mode is made after the required register settings have been made.

In the case of a reset by the cancellation of Deep Software Standby mode, the DPSBYCR.IOKEEP bit is cleared to 0 after the I/O port settings have been made.

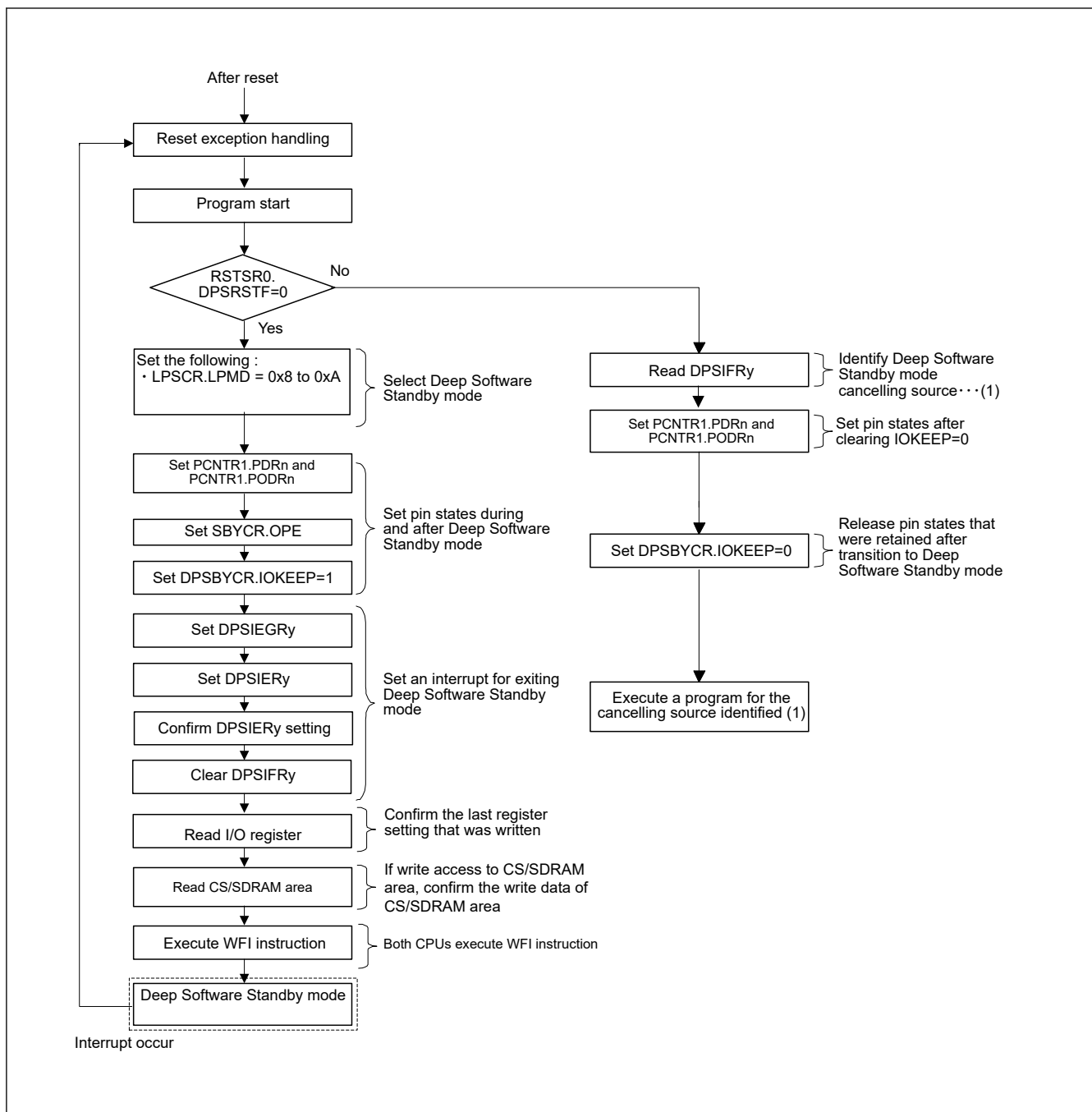


Figure 11.5 Example flowchart to use Deep Software Standby mode

11.7 Voltage Scaling Control

The MCU supports Voltage Scaling (DVFS) control for low power or high performance operation. CM85 TCM and CM85 cache cannot be used during the transition (VSCR.VSCMTSF = 1). Do not access CM85 TCM. Deactivate the CM85 cache or set the cache attribute of the entire address space to Non-cacheable. Figure 11.6 shows the transition of DVFS control.

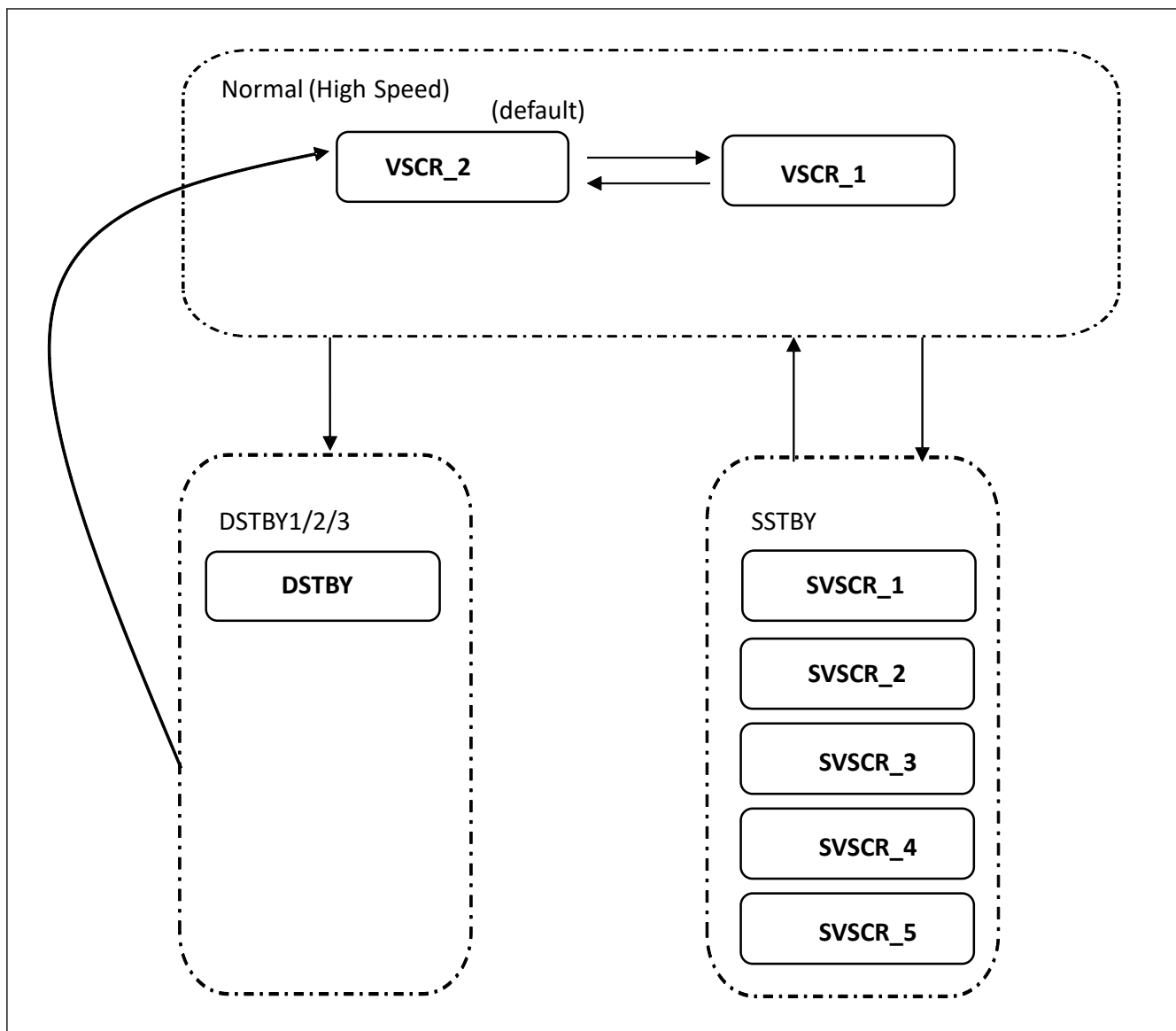


Figure 11.6 Voltage Scaling (DVFS) control for low power or high performance operation

<Normal Mode>

Make sure the operating condition (the frequency range) is always within the specified range before and after switching the VDD in Normal mode. Switch the voltage so that each clock does not exceed the maximum frequency shown in [section 60, Electrical Characteristics](#).

The followings are examples of the procedures for switching the VDD

(1) Switching from VSCR_2 to VSCR_1

1. Set the VSCR.VSCM[2:0] bit to 0x1.
2. Confirm that VSCR.VSCMTSF flag is 0.
3. Set the SCKDIVCR and SCKDIVCR2 to the desired value.

(2) Switching from VSCR_1 to VSCR_2

1. Set the SCKDIVCR and SCKDIVCR2 to the desired value.
2. Set the VSCR.VSCM[2:0] bit to 0x2.
3. Confirm that VSCR.VSCMTSF flag is 0.

(3) In case it is necessary to change the PLL setting when changing the VDD

1. Change the clock source to other than PLL.
2. Stop the PLL.
3. Change the PLL setting.
4. Set the VSCR.VSCM[2:0] bit to the desired value.
5. Confirm that VSCR.VSCMTSF flag is 0.
6. Enable the PLL.
7. Wait the stabilization time for PLL.
8. Change the clock source to PLL.

< Software Standby mode >

- It is possible to transfer to Software Standby mode and to return from Software Standby mode with VSCR set to something other than default at Normal mode.
In Software Standby mode, SVSCR setting is enabled. SVSCR must be set before transitioning to Software Standby mode.
VSCR setting is retained during Software Standby mode and after returning from Software Standby mode.
- It is forbidden to set SVSCR_1 and SS2LP[1:0] = 01b and transition to Software Standby mode.

<Deep Software Standby mode 1/2/3>

- It is possible to transfer to Deep Software Standby mode 1/2/3 and to return from Deep Software Standby mode 1/2/3 with VSCR set to something other than default at Normal mode.
VSCR is initialized when returning from Deep Software Standby mode 1/2/3.

11.8 Usage Notes

11.8.1 Register Access

(1) Do not write to the registers listed below in either of the following conditions:

<Registers>

- All registers with a peripheral name of SYSC.

<Conditions>

- OPCCR.OPCMTSF = 1 (during transition of operating power control mode)
- MENTRYR.MENTRY = 1 (Extra MRAM program mode)
- VSCR.VSCMTSF = 1 (during transition of operating power control mode)
- Time period from executing WFI instructions from both CPUs to returning to Normal mode

(2) Do not write to the registers listed below by DTC or DMAC.

<Registers>

- MSTPCRA
- MSTPCRB
- MSTPCRC
- MSTPCRD
- MSTPCRE

(3) Write access to the registers listed below is invalid when PRCR.PRC1 bit is 0.

<Registers>

- OPCCR
- PGSCR

- PSSTCRn
- PDCTRESWM
- PDRAMSCRn
- SBYCR
- SSCR1
- LPSCR
- DPSBYCR
- DPSIERn
- DPSIFRn
- DPSIEGRn
- PLL1LDOSCR
- PLL2LDOSCR
- HOCOLDOSCR
- LVOCR
- VSCR
- SVSCR
- CPUDSCR
- MWMCR

(4) Write access to the registers listed below is invalid when PRCR.PRC4 bit is 0.

<Registers>

- LPMSAR
- PGCSAR
- DPFSAR
- DPFSAR1
- RSCSAR

11.8.2 I/O Port States

I/O port states in Software Standby mode, Deep Software Standby mode are as same as before entering the modes. Therefore, the supply current is not reduced while output signals are held high.

11.8.3 Module-Stop State of DMAC and DTC

Before writing 1 to MSTPCRA.MSTPA22 or MSTPCRA.MSTPA23, clear the DMAST.DMST bit of the DMAC and the DTCST.DTCST bit of the DTC to 0. For details, see [section 17, DMA Controller \(DMAC\)](#) and [section 18, Data Transfer Controller \(DTC\)](#).

11.8.4 Internal Interrupt Sources

Interrupts do not operate in the module-stop state. If setting module-stop bit while an interrupt request is generated, a CPU interrupt source or a DMAC or DTC startup source cannot be cleared. For this reason, make sure to disable the corresponding interrupts before setting the module-stop bits.

11.8.5 Input Buffer Control by DIRQnE Bit

Setting the DPSIERy.DIRQnE (y = 0 or 1 or 4 or 5, n = 0 to 31) bit to 1 enables the input buffer of the IRQ0-DS to IRQ31-DS pins. Note that, although inputs to these pins are sent to the DPSIFRy.DIRQnF (y = 0 or 1 or 4 or 5, n = 0 to 31) bits, they are not sent to the interrupt controller, peripheral modules, and I/O ports.

11.8.6 Transition to Low Power Modes

Do not enter low power modes (CPU Deep Sleep mode, Software Standby mode, or Deep Software Standby mode) by executing a WFE instruction. It is possible to enter CPU Sleep mode by executing a WFE instruction. In that case, CPU Sleep mode is canceled by the SEV instruction executed by another CPU, or any available interrupt request.

11.8.7 Timing of WFI Instructions

It is possible for the WFI instruction to be executed before I/O register and CS area writes are complete, in which case operation may not be as intended. This can happen if the WFI is placed immediately after a write to a I/O register and CS area. To avoid this problem it is advisable to read back the register and CS area that was written to confirm that the write has completed.

11.8.8 Writing WDT/IWDT Registers by DMAC or DTC

Do not write registers in WDT or IWDT by DMAC or DTC while WDT or IWDT is stopped by entering processor low power mode.

11.8.9 Module Stop Bit Write Timing

It is possible that the access to I/O register may be executed before the corresponding module stop bit write is completed. In this case, the access to I/O register may not proceed as intended. To avoid this issue, before accessing I/O register, read back the module stop bit that was written to confirm that the write completed.

11.8.10 Notes on Transitioning to or Canceling Low Power State

Transitioning to/returning from the low power state requires procedures.

These flows are shown in [Figure 11.7](#) to [Figure 11.12](#). It is recommended to use software to measure the wait time. Be sure to consider the worst-case conditions to ensure that the required wait time elapses. If an interrupt is unavoidably generated during the wait time, retry the measurement after return from the interrupt.

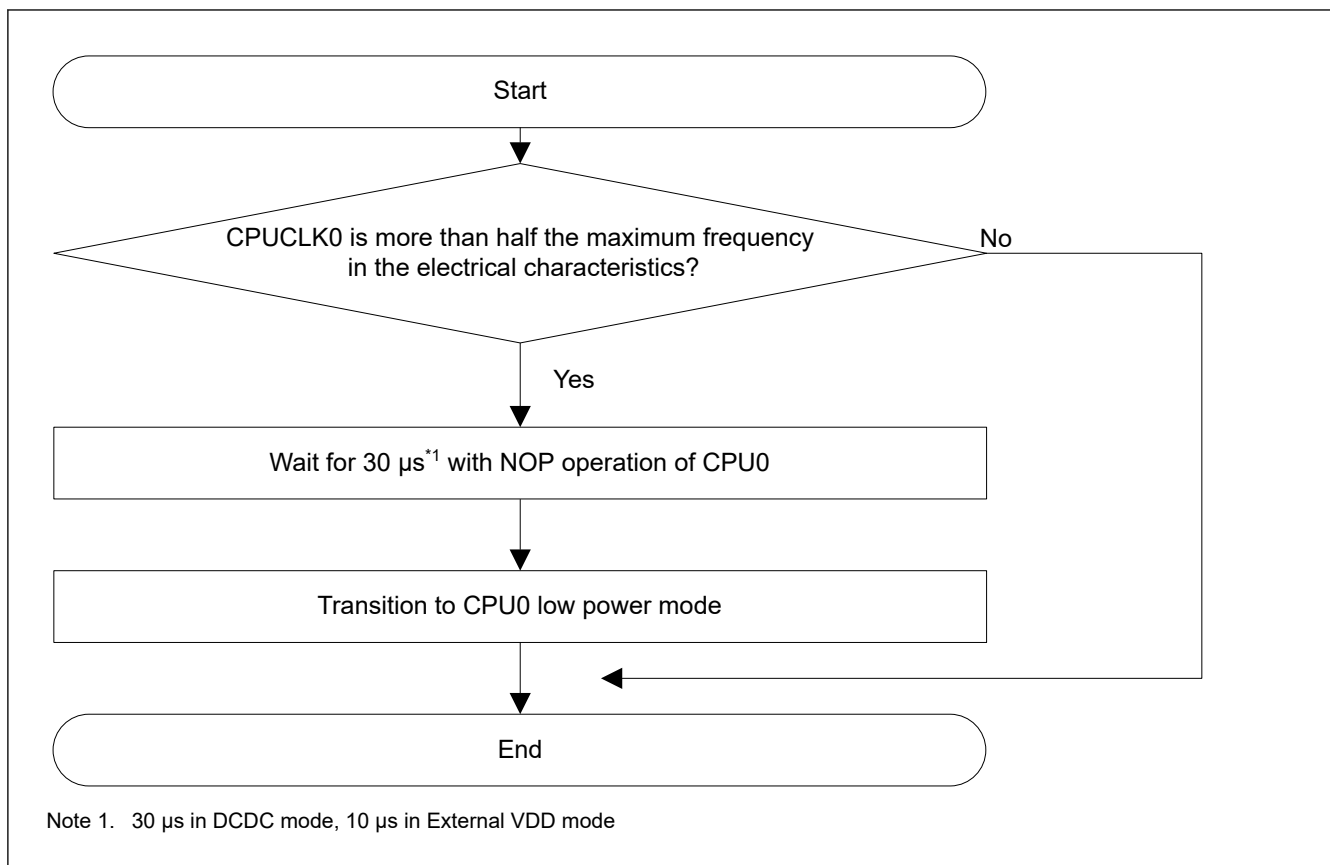


Figure 11.7 Additional flow for the transition to CPU0 Sleep mode, CPU0 Deep Sleep mode with CPUDSCR.PGD0 bit is 1 according to CPUCLK0 frequency

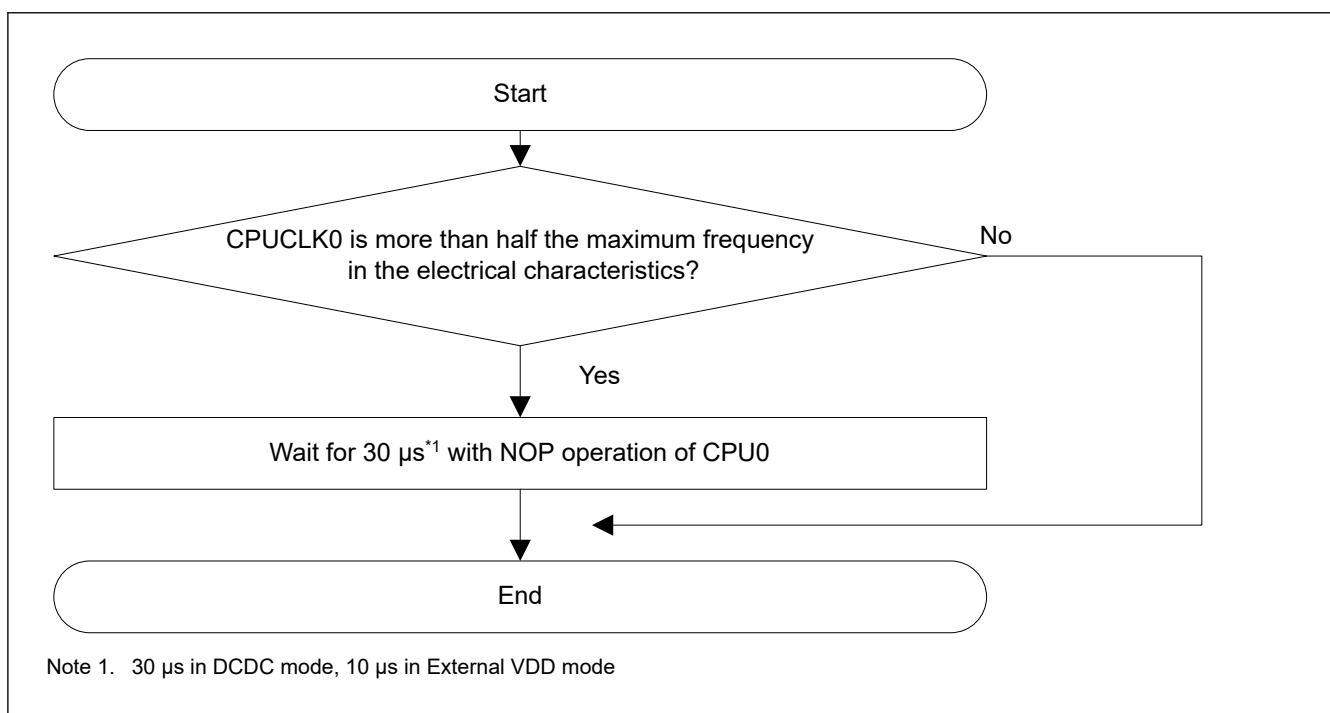


Figure 11.8 Additional flow for the return from CPU0 Sleep mode, CPU0 Deep Sleep mode with CPUDSCR.PGD0 bit is 1 according to CPUCLK0 frequency

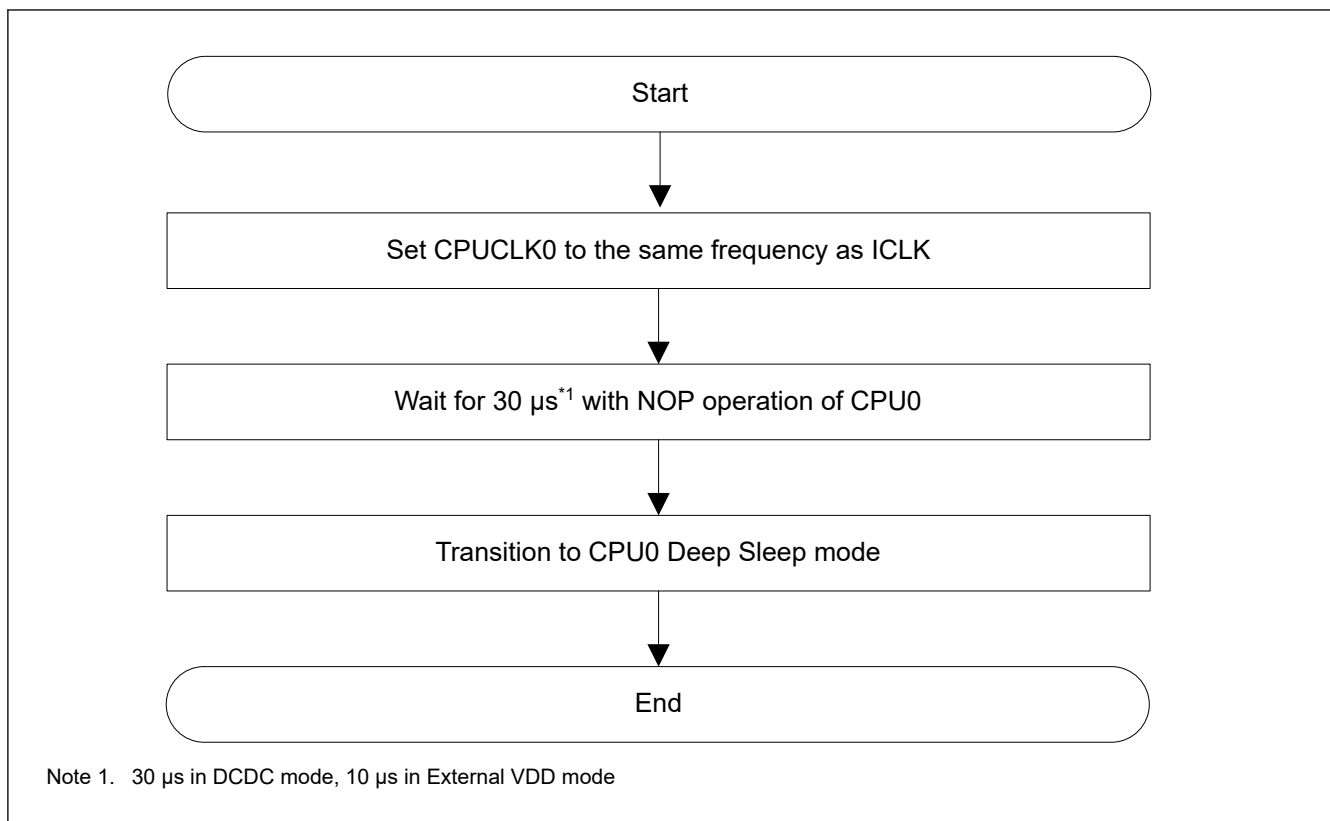


Figure 11.9 Additional flow for the transition to CPU0 Deep Sleep mode with CPUDSCR.PGD0 bit is 0

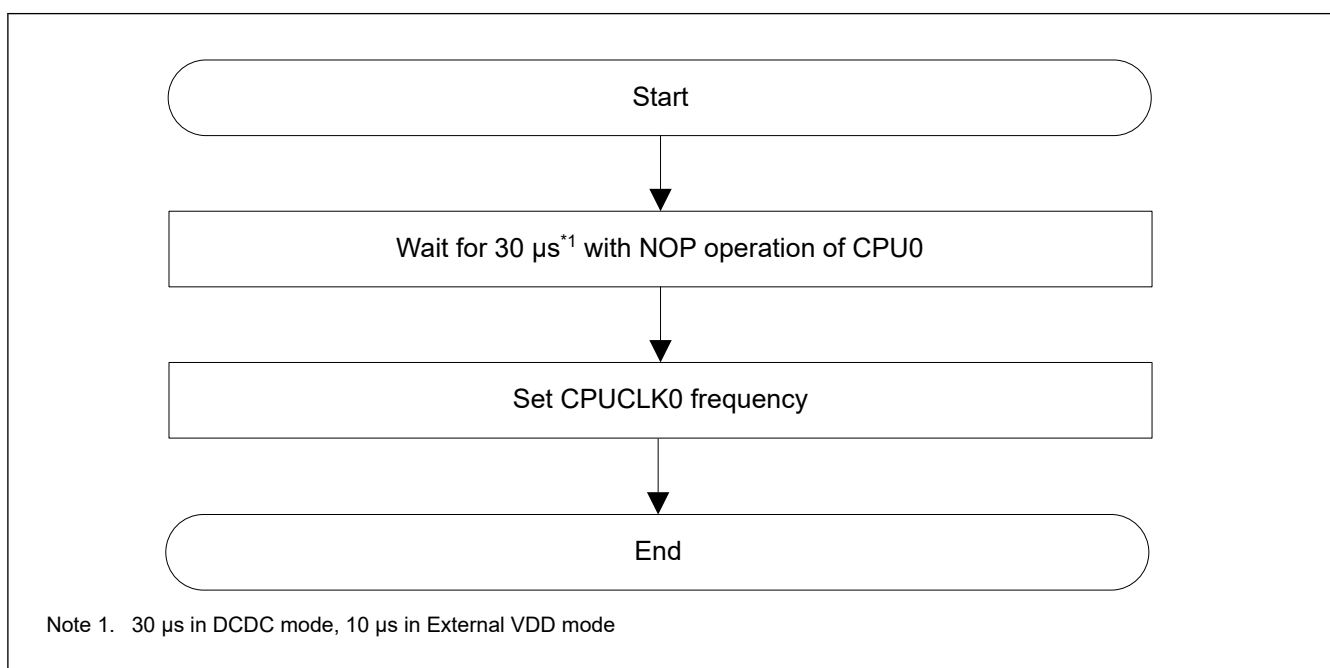
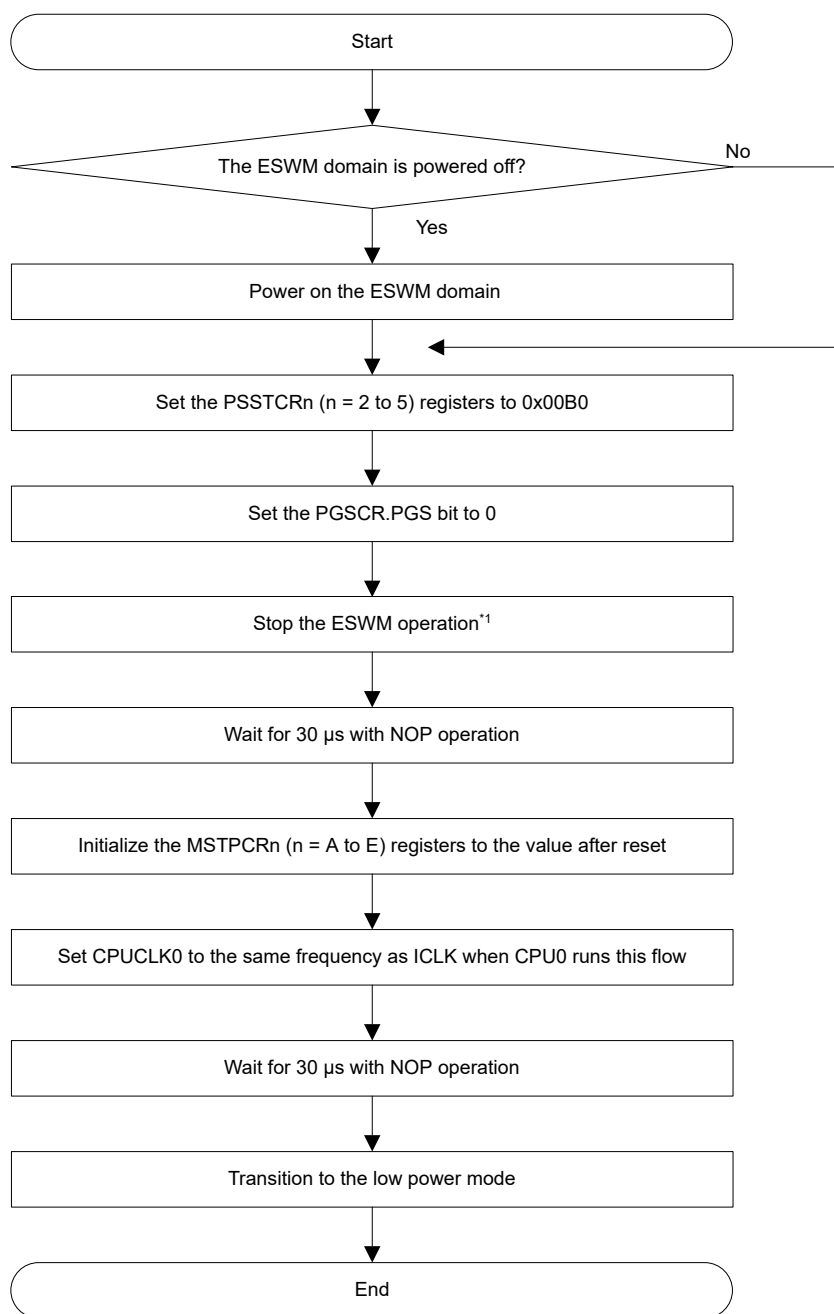


Figure 11.10 Additional flow for the return from CPU0 Deep Sleep mode with CPUDSCR.PGD0 bit is 0 (assuming that CPUCLK0 and ICLK are the same frequency when returning)



Note 1. See the Usage Notes in each chapter for how to stop the operation

Figure 11.11 Additional flow for the transition to Software Standby mode and Deep Software Standby mode 1/2/3

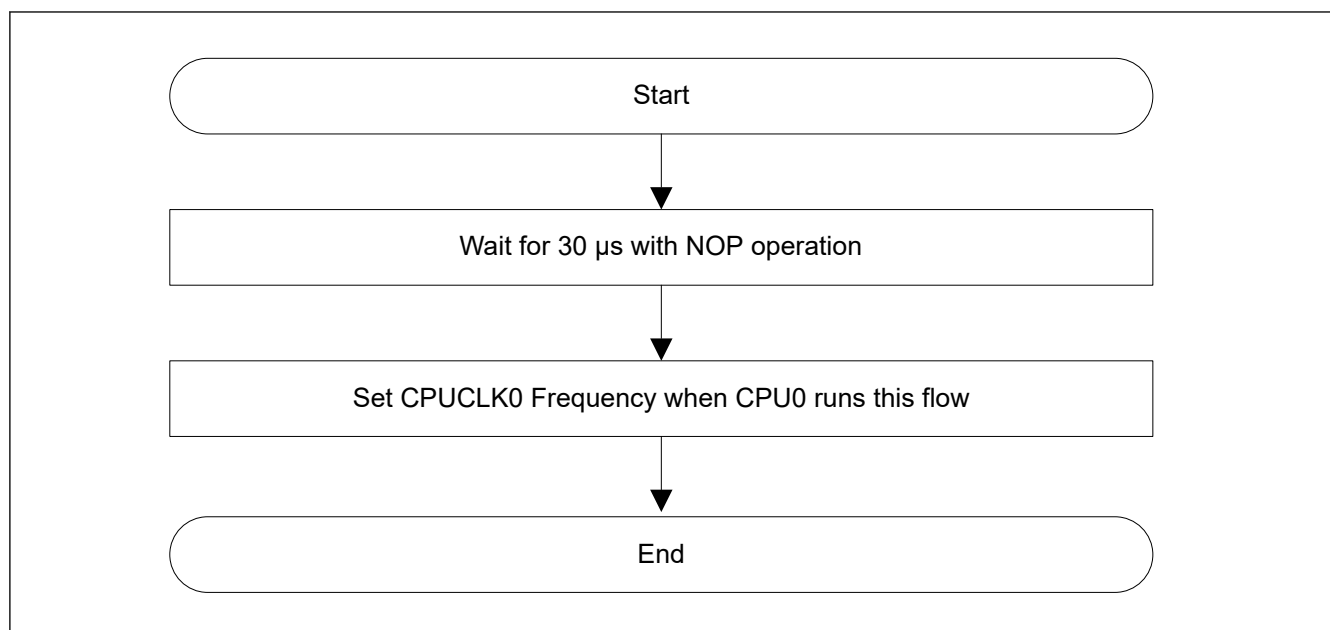


Figure 11.12 Additional flow for the return from Software Standby mode (assuming that CPUCLK0 and ICLK are the same frequency when returning)

11.8.11 Notes about the WFE instruction or Sleep-on-exit function

There are 3 ways to transition to low power modes, WFI instruction, WFE instruction and Sleep-on-exit. When WFE instruction or Sleep-on-exit is used for transition to low power modes, WFI instruction comments written in User's Manual Hardware is applicable to WFE instruction or Sleep-on-exit.